

Occupancy modeling using population statistics and machine learning for urban residential built environment

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ABSTRACT

Occupancy modeling aims to represent the diversity of occupant behavior in buildings, thus, accurate modeling of occupancy is essential for understanding household dynamics and energy-related interactions in residential buildings, as well as for their use in building performance applications. The central contribution of this research is a novel, data-driven methodology for generating high-fidelity occupancy data called CENTUS. Our approach synthesizes official population statistics from ISTAT with nuanced behavioral patterns analyzed using advanced deep learning architectures (LSTM and Transformer). This framework enables the comprehensive, year-long classification of household occupancy across both temporal and non-temporal attributes. Through unified multitask learning that integrates sequential columns with demographic attributes, these models can simultaneously classify multiple occupancy attributes with superior accuracy and broader coverage compared to traditional deterministic and stochastic approaches. Our approach delivers three key advantages: ensures privacy protection through ethically sourced public institutional data; enables cross-national compatibility; and supports flexible scaling from individual residential units to neighborhood-level analysis via multiple modeling strategies including Argmax classification, SoftMax distributions, and temperature-controlled sampling.

1. Introduction

Building performance modeling (BPM) serves as a robust analytical tool for examining performance metrics and generating policy-relevant insights. It enables evaluation across different scales, from individual buildings to neighborhoods, by accounting for building–environment interactions [1,2]. These models typically integrate diverse data sources such as geographic information systems (GIS), building properties, weather conditions, and especially occupancy patterns to capture complex urban dynamics [3]. While disaggregating demand into different systems increases modeling detail [4]. However, the performance gap between predicted and actual outcomes remains a major problem, largely due to poor representation of occupant behavior [5,6].

Residential buildings are particularly important because they account for the largest portion of urban energy use and show highly diverse behavioral patterns. [7]. Therefore, understanding occupancy in residential settings is essential. Occupants directly shape these dynamics through routines and practices [8,9], especially in dense urban areas where variability exist [8]. Social practice theory highlights household

rhythms over time as behaviors occur as interconnected routines rather than isolated actions [9]. Capturing these dynamics requires models that account for both temporal routines (e.g., daily and seasonal schedules) and demographic factors (e.g., age, gender, employment, education). In this way, occupancy modeling becomes essential not just as an input for performance simulations but as a comprehensive approach for realistically representing household behavior and understanding how demographic and temporal factors shape residential activity patterns [10].

1.1. Process of occupancy modeling

Occupancy modeling comprises three key dimensions (Fig. 1): temporal, spatial and component resolution [11]. Temporal resolution refers to the timescales used in analysis, from hourly snapshots to weekly and seasonal trends that capture daily routine fluctuations, e.g., sleeping, cooking [12]. Spatial resolution refers to the physical environment extending from larger urban contexts to individual thermal zones, reflecting how occupants engage with their surroundings at different levels [13]. Lastly, demographic resolution categorizes the

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occupants in terms of demographic factors, e.g. age, employment status [14]. Together, these dimensions reflect occupant activity in built environment and serve as a guide for improvements in energy efficiency and occupant comfort.

Occupancy can be modeled using two different methods that apply to all modeling techniques: space-based and people-based. **Space-based** approaches evaluate how occupant behavior affects thermal zones in terms of adjusting indoor heating, cooling and ventilation features [12]. Key parameters include occupant density, temperature setpoints and power density. **People-based** approaches directly monitor individual behaviors related to HVAC controls and appliance usage, then aggregate these individual actions to generate outcomes comparable to space-based modeling methods [13].

Comprehensive occupancy modeling integrates household demographics with hourly behaviors [8]. Demographic attributes such as age and family structure establish the baseline, while temporal attributes monitor resident movements [15,16]. Behavioral resolutions include equipment usage and temperature control actions that directly related energy demand [17]. Collective household decisions, especially for heating and cooling, further affect total energy use [18]. However, occupant behavior shows substantial variation, as demographically similar groups may display distinct activity patterns and occupancy preferences (Fig. 1). Detailed temporal granularity is therefore essential for precise occupancy representation. This is particularly important given the frequent interaction between residents and building infrastructure.

1.2. Challenges and Limitations in Residential Occupancy Modeling

Occupancy modeling is particularly challenging because occupant behaviors have variations and fluctuating seasonally and across diverse household demographics [19]. There is a greater autonomy of occupants over residential building systems [20–22]. Thus, these reasons make the occupancy component classification especially difficult [23]. Since these behaviors are inherently stochastic and unpredictable, individual-level modeling that incorporates temporal attributes as household interactions is critical for capturing demand variations among residential units [24]. Nevertheless, conventional models frequently rely on oversimplified assumptions, and mostly addressing occupancy attributes independently [5,17]. This practice produces generic patterns that ignore interdependencies and ultimately fail to reflect actual daily routines. Many studies ignore behavioral diversity, creating simplified occupancy models that don't reflect real-world variations [16]. While integrated approaches that combine all the occupancy components can improve accuracy by accounting for these dependencies [17]. This also underlines the need for capturing behavioral complexity with higher resolutions [25,26]. On the another hand, acquiring comprehensive, high-quality occupancy data remains a persistent limitation [27].

Common issues include data scarcity, partial datasets, and limited temporal resolution [28–30], with additional complications from privacy and consent concerns. In addition, occupancy datasets often require extensive preprocessing and careful alignment before creating comprehensive occupancy datasets [1,31].

1.3. Existing Approaches in Occupancy Modeling and Their Deficiencies

Occupancy modeling has advanced from basic fixed schedules [32,33] that assume uniform behavior to more sophisticated methods. As researchers recognized the limitations of fixed profiles, agent-based models (ABM) [34] and Markov chain models [35] emerged. These approaches began incorporating demographic parameters to capture occupant diversity. For example, ABMs treat each occupant as an independent agent, which enables more realistic outcomes of how people interact with building systems [36]. However, while ABMs often struggled with the complexity of multiple occupants, Markov chain methods were noted as limited in representing the full range of occupancy patterns over different time periods [37].

Broadly, these advanced methods fall under four categories: probabilistic, stochastic, statistical, and data-driven strategies [38]. Probabilistic methods introduce randomness to account for less structured actions [39], while stochastic models flexibly represent uncertain and dynamic behaviors [40]. Despite these capabilities, many of these techniques present significant deficiencies for realistic, large-scale applications. Standardized probabilistic profiles, such as those from IEA Annex 66 and 79, overlook critical household-level variations needed for detailed urban analysis [20,41]. Furthermore, richer behavioral simulations like ABM and stochastic models require massive computing power, which severely limits their use beyond individual buildings [42–44].

More recently, data-driven approaches have gained popularity as effective methods for dealing with the complex nature of occupancy modeling [9]. These methods apply machine learning to sensor data, smart meters, and surveys to identify occupancy patterns. They predict occupancy more accurately and capture time-based variations better than conventional methods [45]. These advanced methods have improved the accuracy and adaptability of models, especially for residential buildings [46,47]. However, their implementation faces significant challenges. They frequently target only single variables (e.g., hourly presence) and rarely integrate essential demographic attributes with temporal patterns [48–50]. Furthermore, their reliability is often undermined by issues with data availability [31], consistency [51], and insufficient temporal granularity [51], while combining multiple sources increases methodological complexity.

Similarly, research utilizing Time Use Surveys (TUS) as a primary alternative to sensor data, faces specific structural limitations despite providing realistic behavioral insights. First, a significant data collection

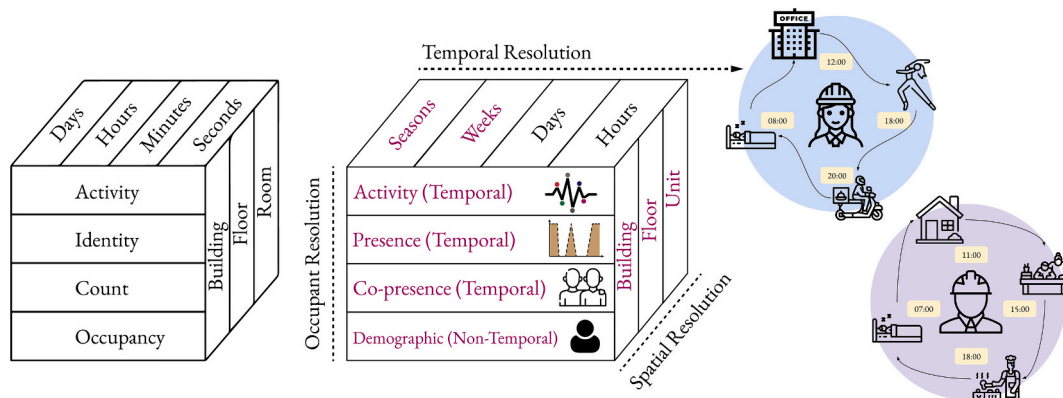


Fig. 1. (left) Dimensions of Occupancy Modeling; (right) examples of temporal activity variations.

gap exists due to the limited duration of records. For instance, TUS diaries typically cover only a 24-hour period, which obscures multi-day patterns and neglects the influence of prior days on current behavior [52]. Second, regarding occupant activity modeling, these sources often fail to account for “co-presence” or shared household activities (e.g., watching TV together or eating dinner). This oversight leads to a tendency toward overestimation. Because models can falsely calculate independent energy loads for activities that are actually shared [53]. Third, integration gaps remain a challenge, particularly the lack of non-energy parameters which is directly related with BEM and UBEM studies. Establishing a meaningful correlation between occupant activities and energy demand requires a deeper integration of social, cultural, and demographic contexts. However, these informations are often absent or disconnected in raw TUS data [54]. Finally, existing methodologies often lack a direct link between physical household characteristics and hourly recordings, limiting the ability to model how specific occupants perform activities based on their unique constraints and properties [55].

A major challenge is the reliance on detailed sensor data, which creates high costs, potential biases, and serious privacy and ethical issues [27]. Consequently, existing methods remain insufficient for generating comprehensive, large-scale occupancy data. As a solution, population-based multi-perspective resources provide a valuable pathway forward. They supply harmonized, nationally representative data with comprehensive component coverage, while simultaneously ensuring ethical collection protocols [56]. These qualities make such datasets well-suited for scalable frameworks. This creates a clear need for new methods that can realistically generate diverse population data while incorporating both non-temporal and temporal attributes.

1.4. Aims and Scope of the Study

This study introduces a scalable framework for generating high-fidelity residential occupancy, specifically designed to address the shortcomings of generalized, resource-intensive, and narrowly scoped previous approaches. The proposed methodology captures detailed occupant behaviors across multiple temporal scales by integrating population statistics using time-series deep learning models (DL), i.e., LSTM and Transformer.

Specifically, this research pursues three primary objectives to address the identified gaps in occupancy modeling:

1. **To synthesize** privacy-preserving, non-temporal population statistics (CENSUS) with detailed behavioral time-use data (TUS) into a unified, high-fidelity occupancy model.
2. **To systematically evaluate** the performance of advanced deep learning architectures (LSTM and Transformer) against conventional stochastic baselines in classifying multi-dimensional occupancy attributes.
3. **To establish** a standardized, scalable framework capable of generating nationally representative occupancy profiles suitable for both individual and neighborhood-level energy simulations.

Two different DL models were trained to show that this methodology can be applied with various time-series algorithms: LSTM, one of the earliest time-series deep learning approaches, and Transformer, a more recent development. The foundation of this framework is the CENTUS dataset, which has all occupancy components consist of non-temporal (residential and demographic attributes) and temporal (“Occupant Activity”, “Presence”, “Co-Presence”) attributes. The dataset is created from a single institutional source, ISTAT by fusing two compatible datasets: CENSUS (Housing and Population) and TUS (Time Use Survey). This single-source approach contrasts with prior works, which often requires complex harmonization of disparate data.

Consequently, this study introduces four key contributions:

1. **Combination of non-temporal attributes with temporal attributes** through deep learning creates a unified dataset. This integration (i) strengthens the classification performance of the models by embedding demographic context into behavioral patterns, and (ii) expands the dataset’s scale and coverage, transforming the limited TUS data into a nationally representative resource of over 200,000 entries.
2. **Exclusive use of TUS data** ensures both demographic attributes and daily diaries are captured without reliance on external datasets.
3. **Standardization of the input data** through the HETUS framework makes the proposed approach globally adaptable across all participating countries.
4. **Flexible configuration of occupancy classification and augmentation** supports applications at multiple scales, from individual residential units to neighborhood-level implementations.

Using this enriched dataset, trained and tuned DL models can directly classify the essential occupancy components, household-related (HH), schedule-related (SCH) etc. Through this pipeline (Fig. 2), publicly available and ethically sourced datasets are systematically translated into multi-dimensional occupancy at the national scale.

2. Datasets

The conversion of population statistics into structured occupancy components consist of several steps as preprocessing, sequential transformation (padding), classification and augmentation, as illustrated in Fig. 7. The approach relies on a comprehensive dataset that integrates building-related attributes, occupant demographics, and occupants’ daily routines, i.e. Italian national datasets collected by Italian National Institute of Statistics (ISTAT), 2011 CENSUS [57] and 2013–2014 TUS [58]. These datasets address ethical concerns by using official public datasets where all households and individuals are anonymously represented through ID numbers, with no personally identifiable information.

2.1. HETUS: Cross-National Compatibility of Population Statistics

HETUS framework is a standardized European framework ensuring cross-national compatibility [60]. All participating countries follow the same diary structure (24 hours divided into 144 ten-minute slots), apply a common hierarchical Activity Coding List (ACL), and record contextual attributes such as location of the activity, and secondary activities. The resident population of the study includes individuals living in private households at their usual residence, identified by shared income or living expenses. Study is targeting participants aged 10 and older, with household surveys conducted face-to-face and individual surveys completed in person, by phone, or via proxy respondents. Each survey delivers three linked files: Household (HH), Individual (IND), and Time Use Diary (TUD), with harmonized attributes such as “Age”, “Household Composition”, and “Employment Status” collected consistently. The common survey design enforces uniform diary formats and data collection periods, while Eurostat ensures quality control through shared manuals (2000, 2008, 2018). Although minor differences remain in coverage, age bands, optional fields, or timing, these can be managed through preprocessing, allowing methodologies developed for one country’s dataset to be directly applied or scaled to others. This structure is enabling robust cross-country analyses.

2.2. Selected Datasets: TUS & CENSUS

Four distinct datasets were used during the flow from model training the inference (Fig. 3). The TUS-Individual and TUS-Daily datasets informed the model training phase, while CENSUS-Housing and CENSUS-Occupant datasets were applied only during inference. The TUS datasets capture household and individual routines in Italy, providing detailed insights into work, leisure, and domestic activities.

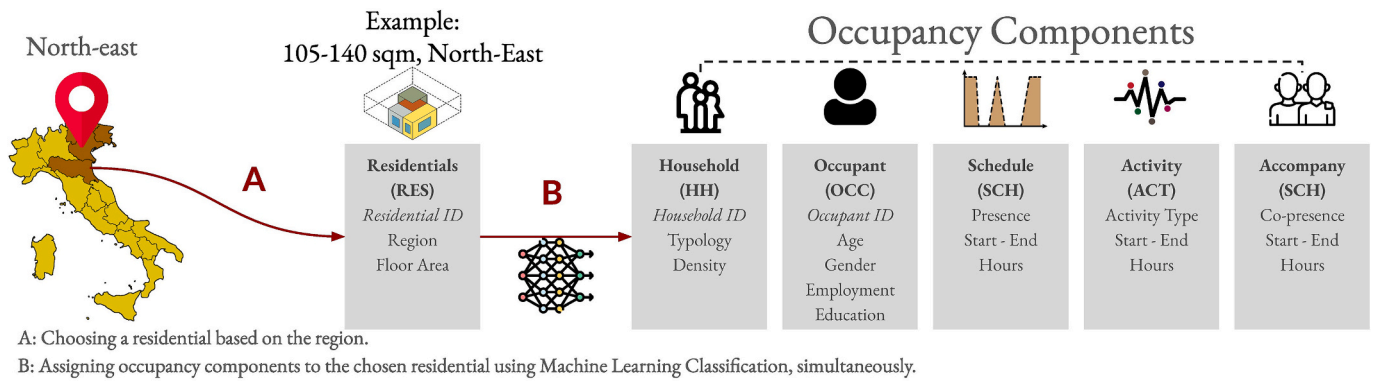


Fig. 2. Assigning Occupancy Components to Residential.

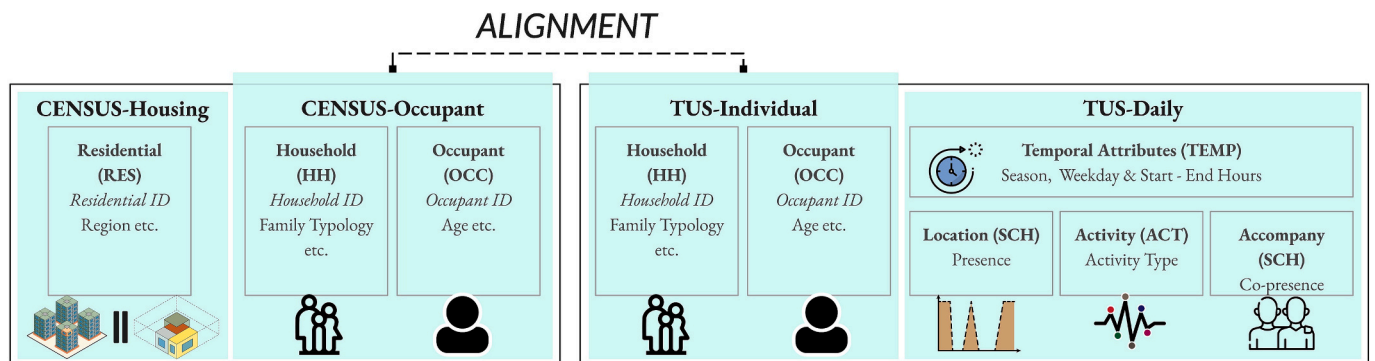


Fig. 3. Integrated Dataset Architecture: CENSUS and TUS with Occupancy Components.

The TUS-Individual subset (2013) includes socio-demographic context from about 34,000 households across 650 municipalities, covering attributes such as “Region”, “Household Size”, “Family Typology”, “Age”, “Education”, and “Employment Status”. The TUS-Daily subset contains temporal attributes for nearly 50,000 individuals, originally collected at 10-minute intervals and standardized to 24-hour sequences (i.e., one daily set) for this study. Key temporal attributes include seasonality (“Months season”), weekly variation (“Week or weekend”), and target variables (“Occupant Activity”, “Presence”, and “Co-Presence”).

The CENSUS provides complementary demographic and housing context, representing approximately 1% of the Italian population sampled across large municipalities and provincial capitals. The

CENSUS-Housing subset describes building and household infrastructure (e.g., “Residential ID”, “Floor Area”, “Room Types”, “Vehicle ownership”), while the CENSUS-Occupant subset details non-temporal attributes, e.g., “Household Location” and frequency of work or study. Together, these datasets align with TUS attributes, enabling their integration and expansion into a unified framework of CENTUS (Table 1). This foundation supports the transformation of static population statistics into detailed occupancy datasets for DL.

2.3. Dimension of Temporal Attributes of Datasets

The temporal resolution of the modeling framework is based on TUS-

Table 1

Content of CENTUS, with the combination of sub-datasets, CENSUS & TUS.

		Data Type	Field Name	Source
Residential	Non-temporal	identifier categorical	“Residential ID” “House Area”, “Room Count”, “Internet Access”, “Car Ownership”, “Home Type”	CENSUS Housing
Household (Demographic)	Non-temporal	binary identifier categorical categorical	“Region” “Household ID” “# of Family Members” “Family Type”, “Nuclear Family Type”	CENSUS Occupant
Occupant (Demographic)	Non-temporal	categorical categorical binary	“Age Class”, “Occupant ID”, “Education”, “Employment”, “Marriage”, “Affiliation”, “Job Type”, “Job Sector” “Occupant Profile Family”, “Occupant Order”	Classified by AI/ML, (3rd step in Fig. 5) CENSUS Occupant
	Temporal	categorical categorical binary binary	“Gender”, “Citizenship”, “Phone Ownership”, “Full/Part Time”, “Permanent/Fixed” “Hour Start Activity”, “Hour End Activity”, “Months season”, “Week or weekend”, “Occupant Activity” (145 unique activities) “Presence” (inside home, outside) “Co-Presence” (accompanied, single)	TUS-Daily Classified & Augmented by AI/DL, (5th step in Fig. 5)

Daily, which specifically consist of occupant daily movements at hourly intervals. This dataset classifies daily sets of occupants according to seasonal and weekly patterns, assigning each occupant a daily set aligned with a particular combination of season and day type (weekday, Saturday, or Sunday). For example, one occupant might have a winter-Sunday daily set, while another follows a summer-Saturday daily set (Fig. 4). Key temporal attributes include activity, location and accompany of activities.

3. Methodology

The methodology consists of the development of a high-resolution occupancy pipeline for residential buildings, culminating in the CENTUS dataset. The primary objective is to use a main DL model, trained on the TUS dataset, to perform two key tasks: 1) classify high-fidelity, hourly temporal attributes and 2) augment these attributes into a full annual representation. Since the target input, the CENSUS dataset, lacks key demographic attributes, a preparatory multi-output neural network (M-NN) is first trained on TUS data to infer these missing columns. This completed CENSUS data is then fed into the main DL model to generate the final dataset. This multi-stage process requires careful coordination of data preparation, model training, and output generation. The following sections detail each stage, beginning with an overview of the CENTUS pipeline, followed by data preprocessing steps, DL model architectures, and probabilistic classification and augmentation methods.

3.1. Development of CENTUS Pipeline

To create a comprehensive occupancy dataset that includes all necessary components for occupancy modeling, we applied the trained DL models to the CENSUS dataset. This dataset provides wider geographical representation than TUS datasets and integrates residential information with related household data. The resulting CENTUS dataset captures both the sequential nature inherent in occupancy modeling and demographic attributes.

The development of the CENTUS involves a systematic pipeline to augment data coverage for all of Italy. It is a structured occupancy dataset suitable for both individual household and neighborhood-level simulations and leveraging probabilistic sampling methods to accurately capture real-world variability among residential units, as shown

in Fig. 5.

1. **Region & Residential Selection (RES):** Choose (manual or random) a specific region to extract residential property details from the CENSUS Housing. Within the selected region, identify and select one or more residential units, specifying their associated physical attributes.
2. **Household Composition Assignment (HH, OCC):** Assign demographic attributes to the chosen residential unit(s). Attributes aligned between datasets are matched using common identifiers, whereas non-aligned attributes are inferred using a multi-output neural network.
3. **Temporal Transformation (TEMP):** Convert non-temporal attributes into temporal attributes, organizing them into 24-hour sequences via padding.
4. **Occupancy Temporal Attribute Classification (SCH, ACT):** Utilize the DL model trained with TUS to classify detailed hourly occupancy data, including temporal attributes, across daily and annual periods. The resulting daily sets and non-temporal attributes are integrated to form the CENTUS.

3.2. Data Preprocessing: Cleaning & Selection & Merging

The ISTAT datasets were originally structured for socio-economic analysis and contain inconsistencies between categories, such as mismatches in household size; therefore, they required extensive preprocessing to be suitable for occupancy modeling (Fig. 1). The preprocessing consists of three main steps: **cleaning**, **selection**, and **merging** (Fig. 7). Cleaning included standardizing headers, removing irrelevant attributes (e.g., “Parking Spaces,” “Potable Water Availability”), and imputing missing values. Selection focused on filtering and simplifying categorical attributes. This ensured alignment with the modeling objectives. The merging integrated demographic, residential, and temporal attributes from four subsets into one dataset. These subsets were combined using common identifiers.

For the TUS datasets, we merged non-temporal attributes from TUS-Individual with temporal attributes from TUS-Daily. Similarly, we cleaned, consolidated and combined CENSUS-Housing and CENSUS-Occupant to provide demographic depth and residential context. Since ISTAT originally compiled these datasets for socio-economic analysis, extensive column filtering was necessary for adaptation to occupancy

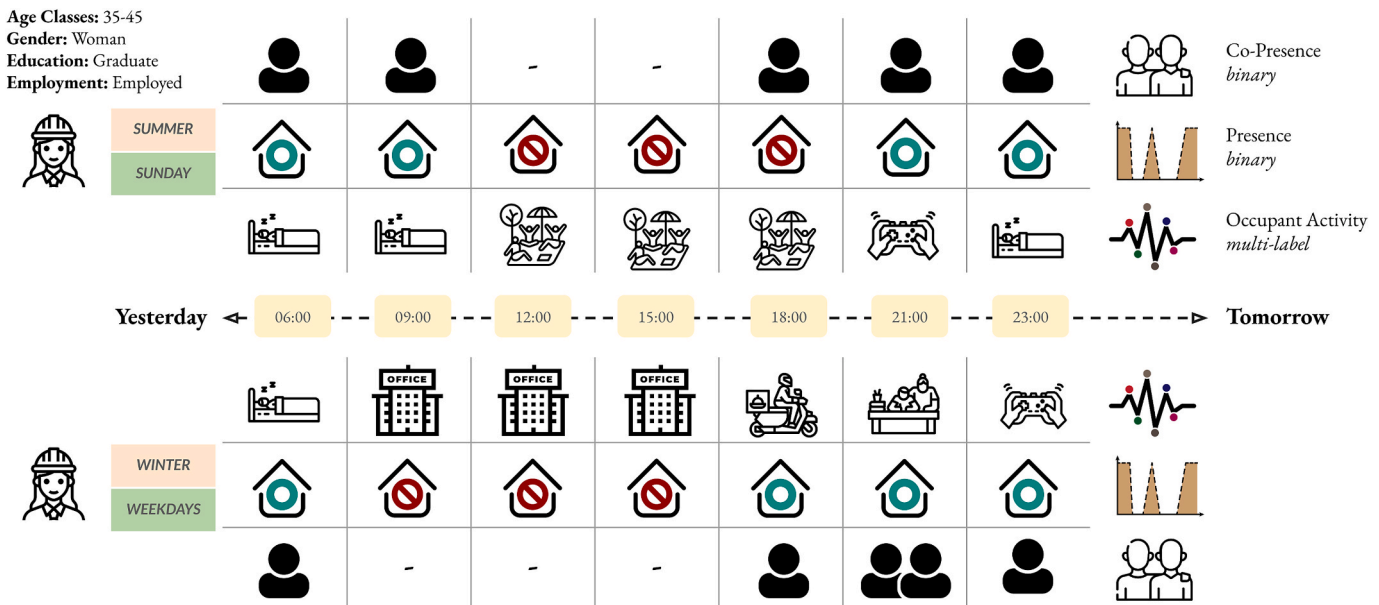


Fig. 4. Dimensions of temporal attribute, daily sets organized by weekday and season.

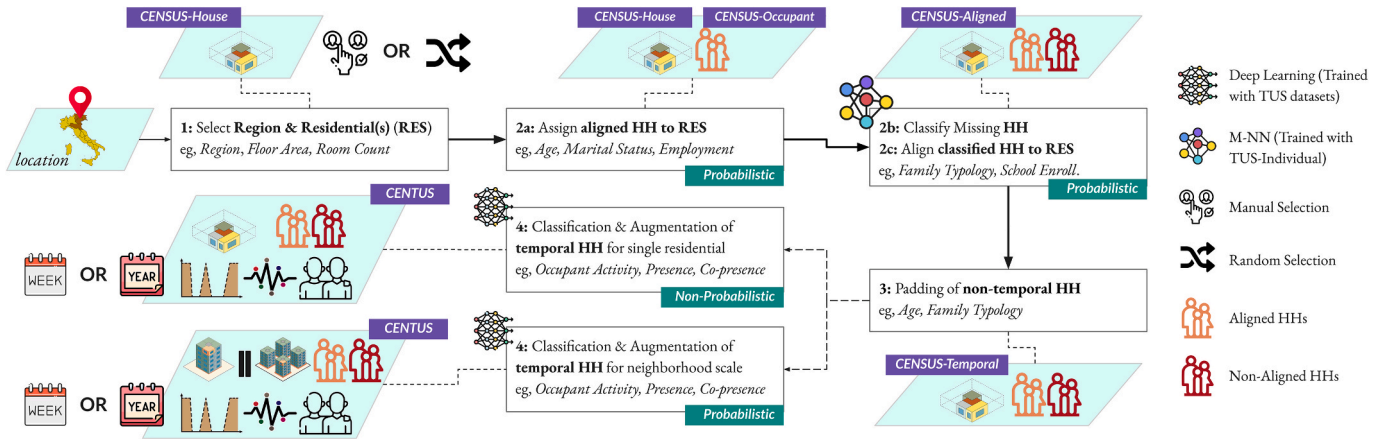


Fig. 5. Four-Stage Pipeline of CENTUS formation.

modeling. Final non-temporal selection prioritized attributes that are most relevant to the three targets. We simplified complex attributes such as “Family Typology” into broader categories to ensure compatibility across datasets. The resulting dataset retained 22 key non-temporal attributes (Table 1).

3.3. Data Preprocessing: Attribute Alignment & Sequential Transformation

The preprocessing stage comprised two primary steps: **attribute alignment** and **sequential transformation**. The CENSUS-Occupant and TUS-Individual datasets shared many common non-temporal attributes, which we harmonized by aligning their categories (Fig. 3). While certain attributes like “Region” and “Gender” were already consistent, others including “Age” and “Marital Status” required harmonization. However, CENSUS-Occupant lacked certain key attributes that existed in TUS, such as “Family Typology,” which is particularly important for accurate temporal classification. To complete these missing columns, we trained a M-NN on TUS-Individual dataset to classify these attributes (Step 2b, Fig. 5; Fig. 6). Initial results showed that M-NN alone did not deliver sufficient performance. Therefore, to enhance classification accuracy, we implemented feature engineering (FE). This involved incorporating household indicators, interaction terms (e.g., “Region” × “Employment Status”), and composite demographic attributes (Appendix B). On the other hand, there is a clear inconsistency for “Co-Presence”. We applied a rule-based correction. By collapsing detailed categories into binary classes and applying logical rules (e.g., assuming married occupants share sleep hours).

During sequential transformation, we normalized hourly sequences using padding and interpolation as uniform 24-hour format for TUS dataset. This captured both long activities, such as sleep, and shorter fragmented tasks. In the CENSUS dataset, which had no temporal

resolution, we structured records to match the same 24-hour format. After these steps, both datasets shared a consistent structure defined by common identifiers. This allowed us to integrate them seamlessly into the DL models. Incorporating these attributes improved the performance of the DL models by enriching the demographic context used in temporal attribute classification. Comprehensive details of attribute alignment are provided in Appendix E1, with the complete category list available in Appendix E2.

3.4. Deep Learning Modeling: Occupancy Classification & Augmentation

DL models combine classification and augmentation within a single pipeline. They firstly classify temporal attributes for a single daily set. Secondly, models augment these sets into full annual representation that capture both seasonal and weekday variations (Fig. 7). The framework can form CENTUS by generating temporal attributes directly from non-temporal CENSUS inputs, removing the need for external sources. Unlike the TUS-augmented dataset generated during training, CENTUS also incorporates residential attributes from the CENSUS-Housing subset, including regional location and dwelling size (Table 1). Although these columns are not required for the DL model itself, their inclusion enhances the richness of the dataset and provides a more complete representation of residential occupancy conditions.

3.5. Deep Learning Model: Preprocessing & Architecture

We prepared all occupancy components for training of DL models during preprocessing phase. We stratified the input dataset to preserve seasonal and weekly distributions across training, validation, and test sets (Appendix C). We excluded the daily sets that do not follow the 24-hour sequence format. The, we converted non-temporal categorical attributes such as “Region” and “Household Size” into integer codes. For

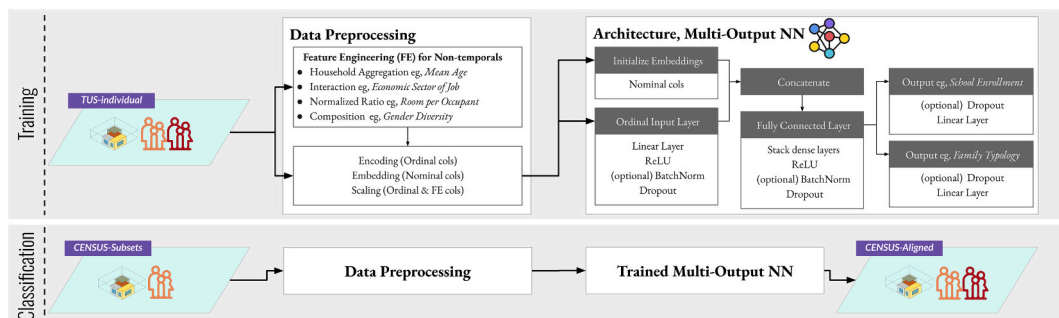


Fig. 6. Machine Learning Modeling for Attribute Alignment.

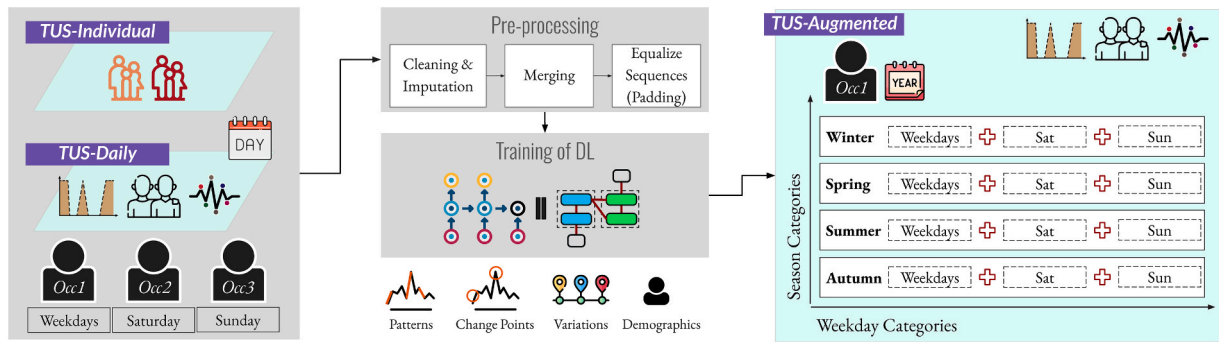


Fig. 7. Flow of Preprocessing & Training of DL Model for Classification and Augmentation.

cyclical attributes like “Hour Start Activity” and “Hour End Activity”, we transformed them with *sine-cosine* functions to represent periodicity. We put numerical attributes into the normalization with standard scaling methods, and lastly, we apply label-encoding for multi-label target column.

During the process, we developed and evaluated two different DL architecture (Fig. 8): Long Short-Term Memory (LSTM) and Transformer models. Both classify temporal attributes as hourly resolution, i.e., $[1, 24, \text{number_of_features}]$. LSTM networks are suitable for tracking of information over long sequences, using gating mechanism, i.e., input, forget, and output gates. This makes them well suited for modeling daily routines that unfold across many hours. On the other hand, Transformer models use attention mechanisms to capture relationships across the entire day, while positional encodings (i.e., using *sine* and *cosine* functions) are added to keep track of sequence order. For training, the models directly process raw logits with suitable loss functions for multi-class (*Cross Entropy Loss*) and binary (*Binary Cross Entropy Loss*) tasks.

Models are based on multitask learning framework that allows the model to perform multiple classifications in a single structure. First, **Conjoint Representation Learning** helps the model capture common patterns across the dataset. This improves generalization to unseen data. Second, learning multiple tasks at once introduces an implicit form of regularization with **Inductive Bias**. This reduces overfitting by preventing the network from focusing on noise. Third, **inter-task knowledge** transfer strengthens the classification of each variable by leveraging the underlying correlations between demographic embeddings and temporal attributes.

3.6. Deep Learning Model: Embedding

Non-temporal categorical columns such as “Region” and “Employment Status” are embedding columns of the model as dense vectors through learnable embedding layers. This enhances demographic representation by providing more informative structures than one-hot

encoding. Temporal attributes (“Months season”, “Week or weekend”) follow the same embedding process, and this enables the model to capture time-dependent relationships directly. The LSTM without embeddings showed stable yet consistently weaker performance, and its loss plateaued at higher levels (Fig. 9). This confirms that categorical embeddings are essential for maximizing model learning. All embedded columns are concatenated with temporal attributes during the training phase. At the last part, the dataset is then reshaped into $[\text{batch_size}, \text{sequence_length}, \text{feature_dim}]$ format for LSTM training, and the Transformer pipeline applies identical preprocessing although Transformer additionally incorporates positional encoding.

3.7. Deep Learning model: Neighborhood-Level Probabilistic Augmentation

Initially, DL framework classifies daily sets for individual buildings by selecting the class with the highest probability (*SoftMax*). This deterministic approach works well at the household level. However, it is insufficient for neighborhood-scale modeling because it produces overly uniform occupant characteristics. In practice, individuals with similar demographics may still follow highly diverse hourly activity patterns. To capture realistic diversity, the CENTUS formation pipeline incorporates probabilistic sampling, as illustrated in Fig. 10. Occupant profiles are generated with models by drawing randomly from the model’s probability distributions, and this introduces variation across temporal attributes. Future applications may also employ Monte Carlo sampling, since the framework is designed to be compatible with such methods. These techniques could further enrich the diversity of daily sets, and the generated occupancy components can be directly used in BEM/UBEM studies.

During inference, the model supports three classification methods, and researchers can select among them depending on the application:

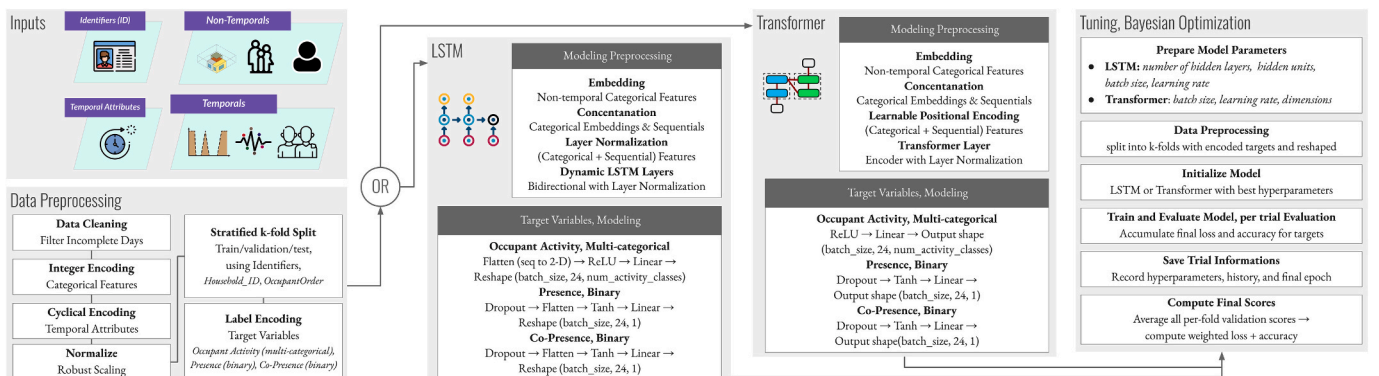


Fig. 8. DL Modeling Architecture & Tuning.

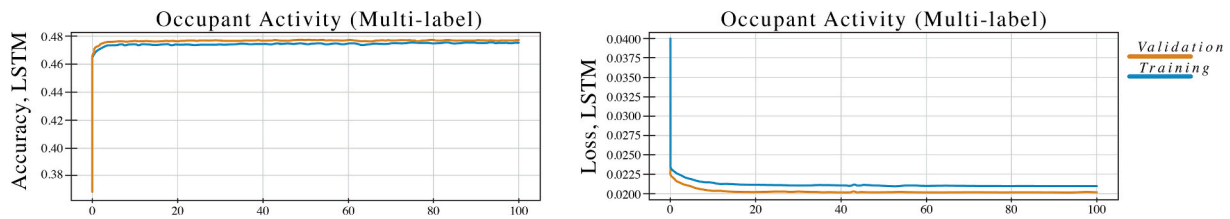


Fig. 9. Learning Performance of Proposed LSTM model without non-temporal embeddings.

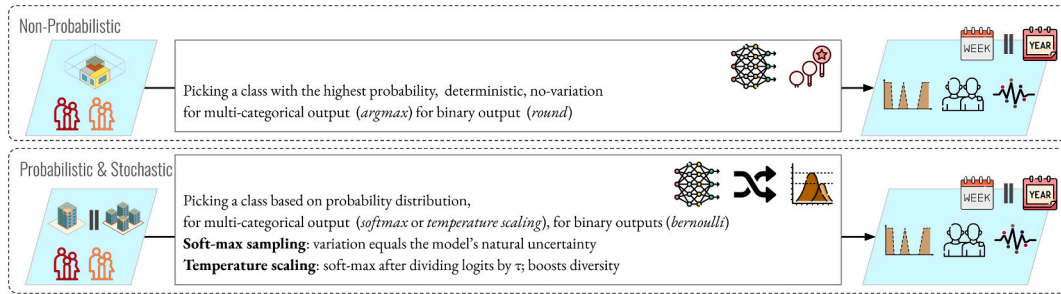


Fig. 10. Neighborhood-level Occupancy Based on Temporal Data Distributions.

1. **Deterministic classification** selects the most probable outcome at each hourly interval using an argmax function. The method applies a fixed 0.5 threshold ($\text{round}(\text{sigmoid}(x))$) to assign binary classes. This produces exact single-household daily sets.
2. **Probabilistic sampling** introduces variation across multiple households. Multi-categorical target “Occupant Activity” are sampled from *SoftMax* distributions, while binary targets are sampled from *Bernoulli*($\text{sigmoid}(x)$) distributions.
3. **Stochastic sampling with temperature and top-k filtering** further increases diversity. The temperature parameter adjusts the distribution: values below 1.0 concentrate probabilities on the most likely outcomes, while values above 1.0 spread them more evenly. *Top-k filtering* selects the k most likely classes and redistributes their probabilities. This method avoids unrealistic outcomes while still providing variation. Typical values are $\text{temperature} = 1.2$ and $k = 5$.

These complementary methods allow occupancy profiles to be generated at different levels of realism and variation. They range from deterministic household daily sets to diverse neighborhood-level patterns.

4. Experiments & Results

This chapter details the experimental results in several key stages. First, a preliminary comparative analysis is presented to quantify the limitations of conventional stochastic models (i.e., Markov Chains) and establish a clear performance baseline. The subsequent sections then explain the core stages of the CENTUS formation including DL training with TUS: (i) the training and tuning (hyper-parameter optimization) of the M-NN model, (ii) a comparative evaluation of the DL architectures for augmentation and classification tasks, and (iii) results of complete occupancy-data augmentation pipeline, which couples temporal attribute classification with the assimilation of household demographics, residential properties, and temporal attributes of TUS datasets.

4.1. Preliminary Analysis: Limitations of Conventional Stochastic Models

Before developing our deep learning framework, we tested whether conventional stochastic methods could meet our research needs. We evaluated first, second, and higher-order Markov chain (MC) models for predicting occupancy patterns. However, these models could only

handle binary presence schedules and cannot perform the simultaneous multi-task classification required for this study. Our experiments showed that adding an “initial state” (IS) improved accuracy—for instance, the higher-order model increased from 0.538 to 0.619 with IS. Nevertheless, this improvement highlighted MC models’ fundamental limitations: they rely on short-term state transitions and cannot capture long-range temporal dependencies in occupant behavior. Table 2 shows these results compared to our deep learning models.

We progressively enhanced the higher-order MC models by adding contextual information (demographic attributes), from no context to partial context (time-based features) to full context (time-based features plus demographic data). Even the most advanced configuration (a higher-order MC model with full context and initial state—achieved only 0.691 accuracy. This was well below the 95% confidence threshold needed for reliable modeling. This analysis shows that even sophisticated Markov chain models cannot meet our needs, confirming the necessity of our deep learning architecture.

4.2. Tuning & Training of the M-NN

We trained the M-NN using two custom-designed pipelines that combine iterative training and validation cycles. First, the *Custom-Preprocessor* integrates feature engineering (FE), encoding, scaling, and embedding transformations into a cohesive, reusable workflow. Second, the *CustomDataset* structures processed data into embedding-ready nominal tensors, scaled ordinal tensors, and specialized target tensors for efficient multi-output classification. Appendix B details the numerous parameters generated through FE. We selected these

Table 2
Markov Chain vs. Deep Learning Models.

Model	Configuration	Task	Accuracy
Markov-Chain (MC)	First Order with IS	Single-tasking, i.e., Presence	0.616
	Second Order with IS		0.619
	High Order with IS		0.691
LSTM	—	Multi-tasking, i.e., Occupant Activity + Presence + Co-Presence	≈0.98
Transformer	—		≈0.98

parameters based on their proven impact on learning performance. Without them, model accuracy plateaued at about 60%. After integrating engineered columns, the M-NN generalized effectively and achieved up to 95% accuracy.

The classification pipeline automates inference by ingesting the CENSUS and classifying household attributes. Table 3 presents an example of results produced by the trained M-NN. During inference, the CustomPreprocessor applies stored transformations to ensure consistent feature representation. The pretrained M-NN then evaluates the transformed dataset, restoring parameters from saved weights (.pth). The model runs in evaluation mode without gradient computations, and it identifies the most probable classes for each attribute using the *argmax* function. We adjusted dropout rates carefully for each output layer: “Family Typology” required a reduced rate (0.0025) due to its complex categories, whereas other family attributes used higher rates (0.05) to optimize performance.

4.3. Comparison of the DL Models

This chapter compares DL architectures in terms of classification of temporal attributes. It examines their performance, applicability, and limitations. Both models use a similar format for handling inputs and outputs. However, they differ in how they handle sequential attributes. The differences include information flow, parallelization, context management, and positional encoding. These are shown in Table 4.

The implementation of models highlights practical differences for temporal occupancy classification (Table 5). Both models process fixed-length sequences and produce identical parallel outputs with common activation (*ReLU*, *Tanh*) and regularization strategies. They differ in input processing. LSTM normalizes embeddings after recurrent layers while Transformers use positional encoding with dual normalization, resulting in LSTMs requiring less memory but Transformers can handle larger batches.

4.3.1. Comparison by Two-step Tuning Process

We tuned hyperparameters using Optuna’s Bayesian optimization with k-fold cross-validation. This approach combines weighted losses from temporal outputs for comprehensive evaluation. It leverages results from earlier trials to guide later ones, which reduces computational cost while averaging performance across folds to prevent overfitting. The tuning process followed two stages (Table 6). In the first stage, we explored wide parameter ranges to identify promising combinations. In the second stage, we refined these candidates to pinpoint optimal settings. Memory usage influenced tuning outcomes: LSTM trials typically reached higher accuracy ($\approx 95\%$) within 30–40 epochs, while Transformer trials required about 90–100 epochs despite faster initial learning.

We applied identical training conditions to both models, including early stopping (patience = 30), learning rate scheduling, and dropout (0.25). Both architectures converged to an embedding size of 50 with zero internal dropout, yet they differed in other hyperparameters. The LSTM achieved its best results with two hidden layers (800 and 600 neurons) and a relatively higher learning rate (1e-3). In contrast, the Transformer performed best with a single hidden layer, a much larger feed-forward dimension (10,240), and a lower learning rate (4e-4). Optimal batch sizes also diverged (256 for LSTM versus 128 for Transformer). These differences reflect architectural distinctions: LSTMs favor deeper recurrent layers, whereas Transformers rely on wider but shallower attention mechanisms.

4.3.2. Training with Best-hyperparameters

The tuned LSTM (Table 6) achieves rapid convergence, and it surpass ≈ 0.90 accuracy for the multi-label task (“Occupant Activity”) and ≈ 0.96 for the binary tasks (“Presence” and “Co-Presence”) within 30 epochs (Fig. 11). Around epoch 120 it plateaus at ≈ 0.960 (multi-label) and ≈ 0.98 (binaries) on the validation set, only 0.5–1 percentage higher

Table 3
Randomly Selected Household, Classified Outputs with trained M-NN.

INPUTS		Members of Household		Region	House Area	# Room Count	Internet Access	# Family Members	Simple Family Type		**Home Ownership	Car
Gender	occupant 1	Age Class	Marital Status	Education	Affiliation	Employment	Full/Part Time	Employment Type	Job Type	Job Sector	Citizenship	Phone Owner.
	occupant 2	25-34	Single	Diploma	Children	Student	Unemployed	Not included	Unemployed	Unemployment	Italian	Yes
	occupant 3	65-74	Married	Diploma	Reference Person	Employed	Full Time	Not included	Entrepreneur	Services	Italian	Yes
Female		55-64	Married	Non-qualified	Spouse	Retired	Unemployed	Not included	Unemployed	Unemployment	Italian	Yes
IDENTIFIERS		*Residential ID	*Household ID	Occupant ID	OUTPUTS	Family Type	Nuclear Family Type		Occupant Profile		Occupant Role Family	
		7,663,430	7,663,430	3	Married couple with children without isolated	Married couple with children without isolated	Couple with children	Couple with children	Child	Nucleus leader	Core Family	
		7,663,430	7,663,430	1	Married couple with children without isolated	Married couple with children without isolated	Couple with children	Couple with children			Core Family	
		7,663,430	7,663,430	2	Married couple with children without isolated	Married couple with children without isolated	Couple with children	Couple with children	Spouse		Core Family	

Table 4
Architectural Model Comparison for Temporal Attribute Classification.

Modeling	LSTM Model	Transformer Model
Architecture & Processing	Sequential processing through bidirectional LSTM, number of hidden layers $\times$ (Bi-LSTM, LayerNorm, Dropout)	Parallel processing using multi-head self-attention (n_{head}), number of hidden layers $\times$ (Multi-head Attention, Feed Forward Network (FFN), residual)
Temporal Resolution	Implicitly captured via hidden state propagation	Explicitly encoded through learnable positional embeddings
Context Handling	Memory-efficient for shorter sequences	Better at modeling long-range dependencies but less memory-efficient
Constraints	No special constraints	Input dimension must be divisible by number of attention heads (n_{head})
Computational Properties	Linear complexity $O(L \cdot d^2)$, Limited parallelization, Slower training with risk of vanishing gradients despite gates	Quadratic complexity $O(L^2 \cdot d)$, Parallelizable, Scales with sequence length, requires careful n_{head} choice and more RAM

Table 5
Comparison of DL Model for Implementation Differences.

Implementation	LSTM Model	Transformer Model
Input Processing	Concatenated (Embeddings categorical non-temporals & temporals) $\rightarrow$ Norm $\rightarrow$ LSTM layers	Concatenated (Embeddings categorical non-temporals & temporals) $\rightarrow$ Positional Encoding $\rightarrow$ Norm $\rightarrow$ Transformer encoder layer
Normalization	LayerNorm after each LSTM layer	LayerNorm before and after transformer encoder blocks
Regularization	On output heads (0.25 for “Presence”, “Co-presence”)	On output heads (0.25 for “Presence”, “Co-presence”)
Hyperparameters	hidden unit sizes, number of LSTM layers	number of attention heads, feed-forward dimension, number of encoder layers
Parameter Sharing	Same LSTM weights across timesteps	Attention & FFN weights shared across positions within layers
Memory Usage	Lower activations, smaller model for same hidden size, thus, lower memory footprint	Higher (stores attention maps) but supports larger batch sizes

than training values (0.948 and ≈ 0.98). It yields overall validation and training accuracies of 0.976 and ≈ 0.970 . Thus, it indicates strong generalization with minimal over-fitting. Cross-entropy loss follows the

Table 6
Hyper-Parameter Search Ranges and Optimal Settings for LSTM and Transformer.

LSTM	# hidden layers	# of neurons, hidden units	dropout, LSTM	batch size	learning rate	embedding size	
initial trials	[2, 3, 5, 7]	[100, 300, 500, 700, 900]	[0, 0.25]	[24, 96, 256, 512, 1024]	[1e-2, 1e-3, 1e-5, 1e-6]	[25, 50, 100, 150]	
refinement trials	[2,3]	[500, 600, 700, 800, 900, 1000]	[0]	[64, 128, 256, 384, 512]	[1e-3, 9e-4, 7e-4, 5e-4]	[50]	
best hyper-parameters	[2]	{layer0: 800, layer1: 600}	[0]	[256]	[1e-3]	[50]	
Transformer	# hidden layers	dimension, feed forward	dropout, Trans.	batch size	learning rate	embedding size	n_head
initial trials	[1, 2, 3]	[4096, 6144, 8192]	[0, 0.25]	[24, 96, 256, 512, 1024]	[5e-3, 1e-3, 5e-4, 1e-4]	[25, 50, 100, 150]	[2, 4, 8]
refinement trials	[1]	[7168, 8192, 9216, 10240]	[0]	[96, 128, 256]	[8e-4, 6e-4, 4e-4, 2e-4]	[50]	[4]
best hyper-parameters	[1]	[10240]	[0]	[128]	[4e-4]	[50]	[4]
Constants	dropout for outputs: dropout_activity= 0, dropout_presence= 0.25, dropout_presence= 0.25, training: Early stop (Patience = 30, min_delta = learning rate), Learning Scheduler (Patience = 30, Factor = 0.95)						

same pattern as combined training loss drops to ≈ 0.060 and validation loss to ≈ 0.050 , while per-task losses converge to 0.104 and ≈ 0.042 in training dataset and ≈ 0.080 and ≈ 0.030 in validation dataset.

The tuned Transformer (Table 6) model improved quickly, as it can be seen in Fig. 12. In just 40 training rounds, its accuracy jumped from 50% to about 90% for “Occupant Activity” classification, and from about 75% to about 96% for binary tasks. Then, it continued improving slowly until training stopped at round 191. These were only slightly better than the training results (about 94.8% for multi-label, 98% for binaries), showing the model wasn't just memorizing dataset. In parallel, error rates dropped dramatically at first (from 2.6 to 0.15 for multi-label; from 0.5 to 0.06 for binaries) before leveling off at about 0.056 during training and 0.049 during testing. The specific test error rates were 0.087 for multi-label and about 0.03 for binaries.

The Reduce-on-Plateau scheduler automatically lowers the learning rate when the model stops improving, this helps to reach smaller improvements. Transformer's accuracy graph shows a zigzag pattern because DL models use different settings. Transformer uses a lower learning rate (0.0004), larger internal units (10,240), and smaller batch size (128), which causes it to hit improvement plateaus quickly. When this happens, the learning rate gets reduced often, creating those visible up-and-down fluctuations. The LSTM uses a higher learning rate (0.001), moderate-sized processing layers (800, 600 units), and larger batch size (256). This allows it to keep making small improvements before each patience check, resulting in a smoother curve.

4.3.3. Comparison of Classification Performance of Activities

“Occupant Activity” presents greater classification challenges, so we evaluated models beyond overall accuracy. We compared LSTM and Transformer results for both the most frequent and least frequent activities based on classification metrics, as illustrated in Fig. 13 and Fig. 14. Recall measures the share of correctly identified actual activities, Precision indicates classification accuracy, and F1-score provides a balanced measure of overall performance. We calculated these metrics at each hourly time step. This allowed us to evaluate how well the models' classified activities throughout the day, rather than only in aggregate. Both models were trained with tuned hyperparameters to ensure a fair comparison. We also assessed all unique activities in Appendix D.

Models perform well across most activity categories (Fig. 13), including those with relatively low representation, but show a notable decline for extremely rare cases (e.g., Work Break, $\approx 0.0002\%$) (Fig. 14). This reduced performance is not due to the models' inability to learn such activities but rather reflects the effect of intentional generalization, which constrains accuracy to 95% to prevent memorization. Within this framework, the Transformer exhibits marginally better performance

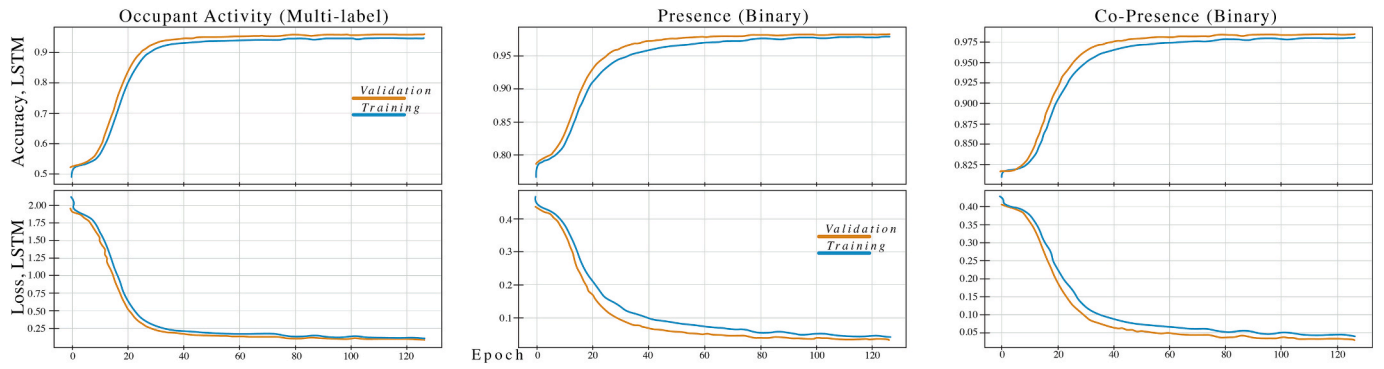


Fig. 11. Training with best-hyperparameters after tuning, LSTM.

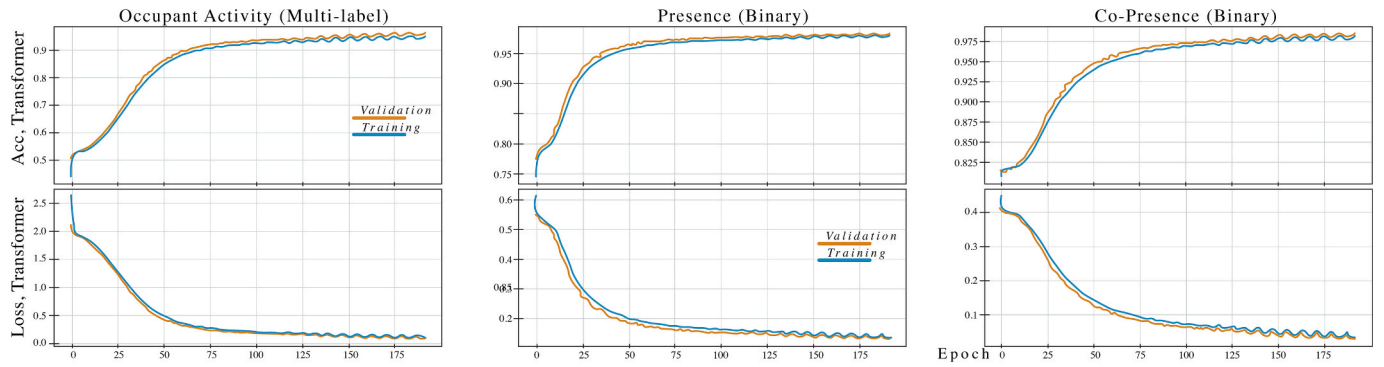


Fig. 12. Training with best-hyperparameters after tuning, Transformer.

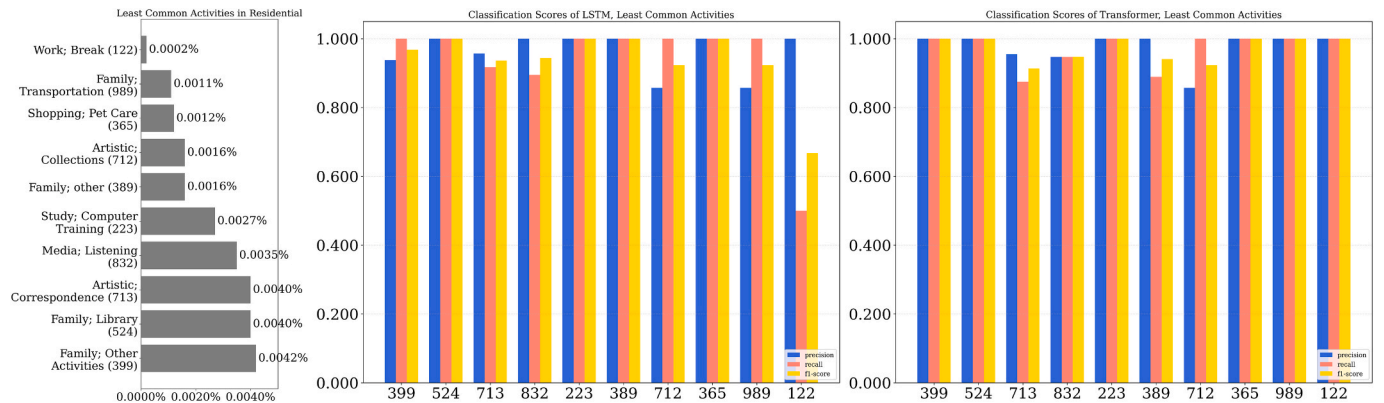


Fig. 13. Occurrence of Activities for LSTM and Transformer, Scores for Most Common Activities.

than the LSTM on the least frequent categories, maintaining stronger classification stability under generalization.

Both models achieve consistently high scores across these common activities. Minor variations between *Precision* and *Recall* suggest slight differences in model tendencies for false positives or missed events, yet overall, both architectures perform similarly well, indicating strong generalization capabilities. Performance stays strong for these activities even with lower occurrence rates.

4.3.4. Classification Performance Comparison with Confusion Matrix

The **confusion matrix** is a critical for evaluating classification performance in multi-label time series tasks that offers immediate insight into which activities or time steps are most frequently misclassified, distinguishing between errors of omission (false negatives) and commission (false positives). This granular feedback supports targeted tuning of loss weights, thresholds, sampling strategies, or model

architecture, allowing for precise optimization rather than trial-and-error.

Confusion matrices demonstrate strong performance for both common (Fig. 15) and less frequent activities, confirming both models' suitability for detailed residential activity classification. LSTM achieves notably high recall rates, such as 99% for "Sleep (11)" 97%, for "Eat/Drink (21)" 95%, for "Media; general (31)" and at least 96% for the other main activity categories. Similarly, Transformer matches these results, achieving recall rates near 99% for "Sleep (11)" and "Work (111)," 97–98% for "Eat/Drink (21)" and "Personal Care (311)," and 96–97% for "House Cleaning (321)" and "Rest (531)."

Diagonal-masked confusion matrices reveal off-diagonal errors below 1% for both models (Fig. 16), with Transformer marginally reducing systematic misclassifications compared to LSTM. The most notable systematic confusion occurs in "Media; general (31)" being misclassified as "Sleep (11)" (2.2% for LSTM, 2.3% for Transformer).

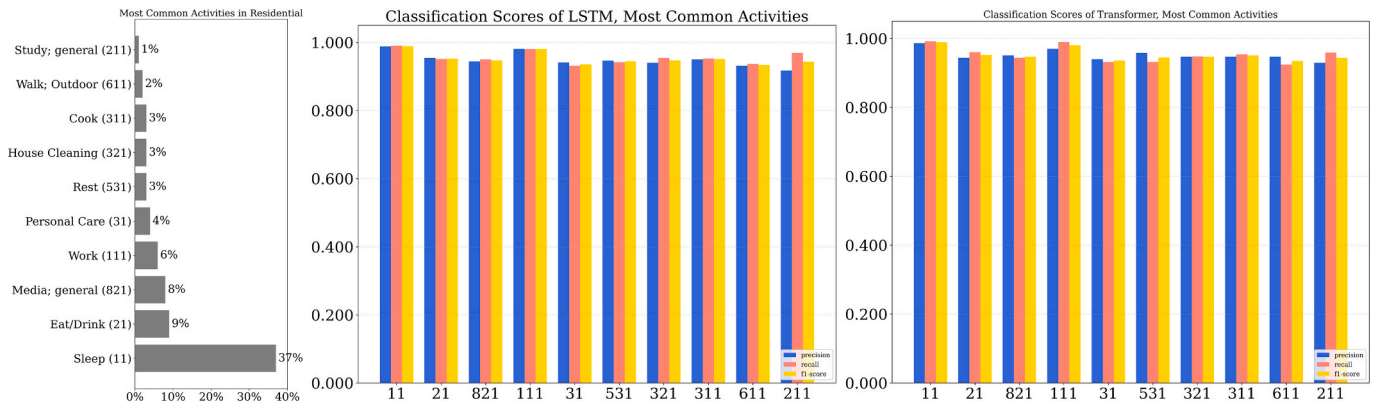


Fig. 14. Occurrence of Activities for LSTM and Transformer, Scores for Least Common Activities.

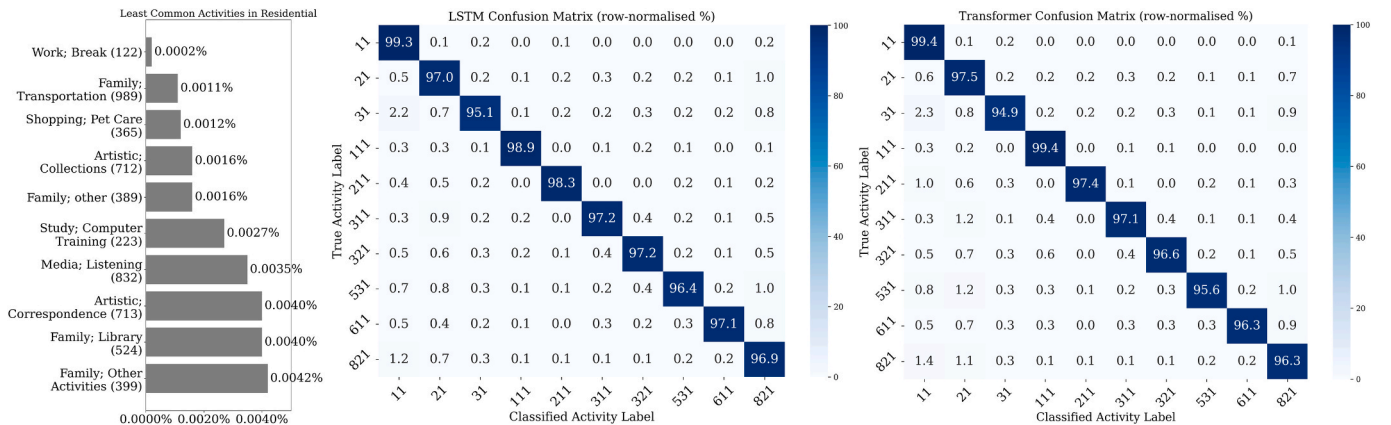


Fig. 15. Confusion Matrix for LSTM and Transformer, Most Common Activities.

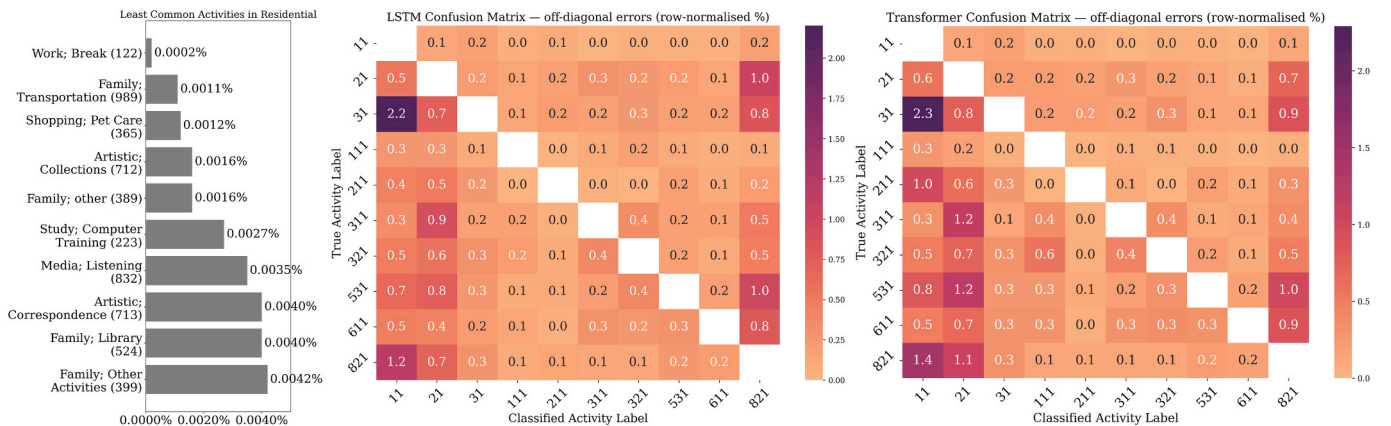


Fig. 16. Confusion Matrix off-diagonal error for LSTM and Transformer, Most Common Activities.

Other minor chains include “Eat/Drink (21)” and “Personal Care (31)” being confused with “Sleep (11)” and each other, but these rarely exceed 1%. Both models effectively distinguish main activities despite small errors, but confusion between similar categories (11, 21, 31) indicates room for improvement via feature engineering or adjusted loss weighting.

4.4. Augmentation of Temporal Attributes

Fig. 17 and Fig. 18 show augmented occupancy daily sets for two adults (married male and female, ages 45-54) from a four-person

household (175771100) in central Italy. The household is the production of LSTM's *probabilistic* inference mode for enhanced variability, as detailed in 3.3. The remaining two members' daily set are provided in Appendix A. The yellow daily set shows the baseline classification used for augmentation. The model first classifies a single daily set, then augments it, ensuring all subsequent daily sets maintain consistency with this baseline for realistic patterns. The occupancy daily sets use color-coding to show “Co-Presence” daily sets: orange lines indicate activities with others (family or external), while blue lines represent solitary activities. This reveals the married couple consistently shares “Sleep (11)” activities (orange) but conducts “Work (111)”

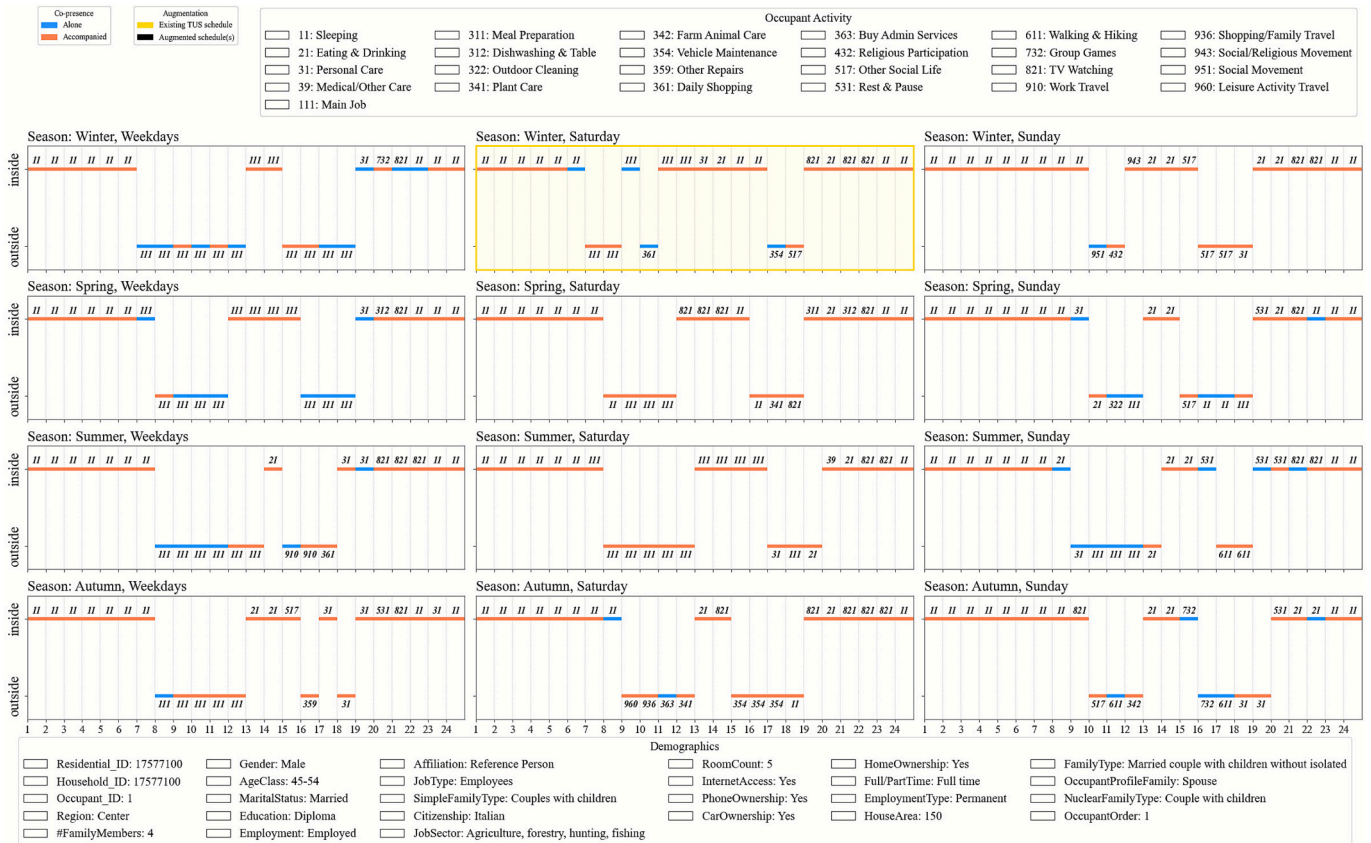


Fig. 17. Father-Married; 4-person Household, Augmented Output Variables.

independently during daytime (blue).

The male occupant maintains a structured routine (Fig. 17): “Sleep (11)” in early morning and late evening, “Work (111)” during conventional hours (8:00–17:00) on weekdays, and evening leisure activities including “Watching TV (821)”, “Eat/Drink (21)”, and “Rest (531)”. Seasonal and weekend shifts slightly change “Social/Religious Events (432)” and “Travel for Work (910)”, and that proves the model adapts behavior realistically from demographic data. A notable observation is that occupant activities vary more than binary tasks, staying consistent daily throughout the year but changing notably on Sundays due to no work.

The retired female occupant maintains a consistent, year-round routine primarily focused on domestic tasks and personal care (Fig. 18). Her core daily activities include “House Cleaning (321)”, “Meal Preparation (311)”, “Dishwashing & Table (312)”, and “Ironing (332)”, complemented by leisure activities such as “Watching TV (821)” and “Rest (531)”. Her daily set remains stable throughout the year, showing minor variations in activity types and durations instead of distinct weekday-to-weekend differences. This reflects the expected lower variability compared to her employed spouse due to employment status. Overall, these findings highlight the CENTUS formation pipeline’s effectiveness in realistically generating occupancy patterns purely from demographic attributes, and that reinforces its suitability for data-driven occupancy modeling applications.

4.5. Case Study: Integration of Classified Occupancy to the BEM

To quantify the impact of high-fidelity occupancy schedules on building energy performance, we took BEM simulations that compared standardized ASHRAE baseline schedules (uniform, continuous occupancy) with “modified” empirically-derived schedules from the 2013 Italian TUS-dataset. These modified schedules include hourly data for

presence, classified metabolic activity, lighting interactions, and equipment usage. Simulations were conducted on ASHRAE 90.1 prototype high-rise and mid-rise apartment buildings in two climate zones: 3A (warm-humid) and 4A (mixed-humid). To ensure methodological coherence between occupancy behavior and building physics, we design the BEM setup as post-2005 construction standards, focusing on the 2010-2015 reference period.

Temporal analysis shows classified occupancy significantly alters daily energy profiles, especially in extreme seasons when active HVAC dominates the energy balance, as presented in Fig. 19. Modified simulations show substantially higher winter heating demand, reflecting a “thermodynamic penalty” from intermittent occupancy. Conversely, modified scenarios reduce summer cooling demand, as occupant absence during peak solar gain hours lowers the instantaneous cooling burden compared to continuous default occupancy assumptions.

Integrating realistic occupancy modeling reduces overall energy use intensity by 5–7% and significantly improves end-use distribution predictions, as shown in Fig. 20. Modified simulations (64–82 kWh/m²) align well with validated Italian residential data (2010–2015). This shows superior agreement across all categories compared to default scenarios. Specifically, modified heating intensities (13–25 kWh/m²) match benchmarks (10–30 kWh/m²). This corrects the systematic underestimation by default models (1.6–17 kWh/m²) which fail to capture occupancy-driven internal gains. For cooling in Zone 3A, modified results (14–17 kWh/m²) fall within the documented range (15–35 kWh/m²), correcting default overprediction (19–21 kWh/m²). Equipment energy also saw a 16–17% reduction in modified scenarios (33–34 kWh/m²). Crucially, realistic modeling successfully captures the “thermodynamic inversion” in Mediterranean architecture (e.g., Zone 3A) where cooling approaches or exceeds heating loads (14–17 kWh/m² vs. 1.6–13 kWh/m²). This confirms that occupancy-integrated, temporally calibrated modeling accurately captures the complex behavioral and

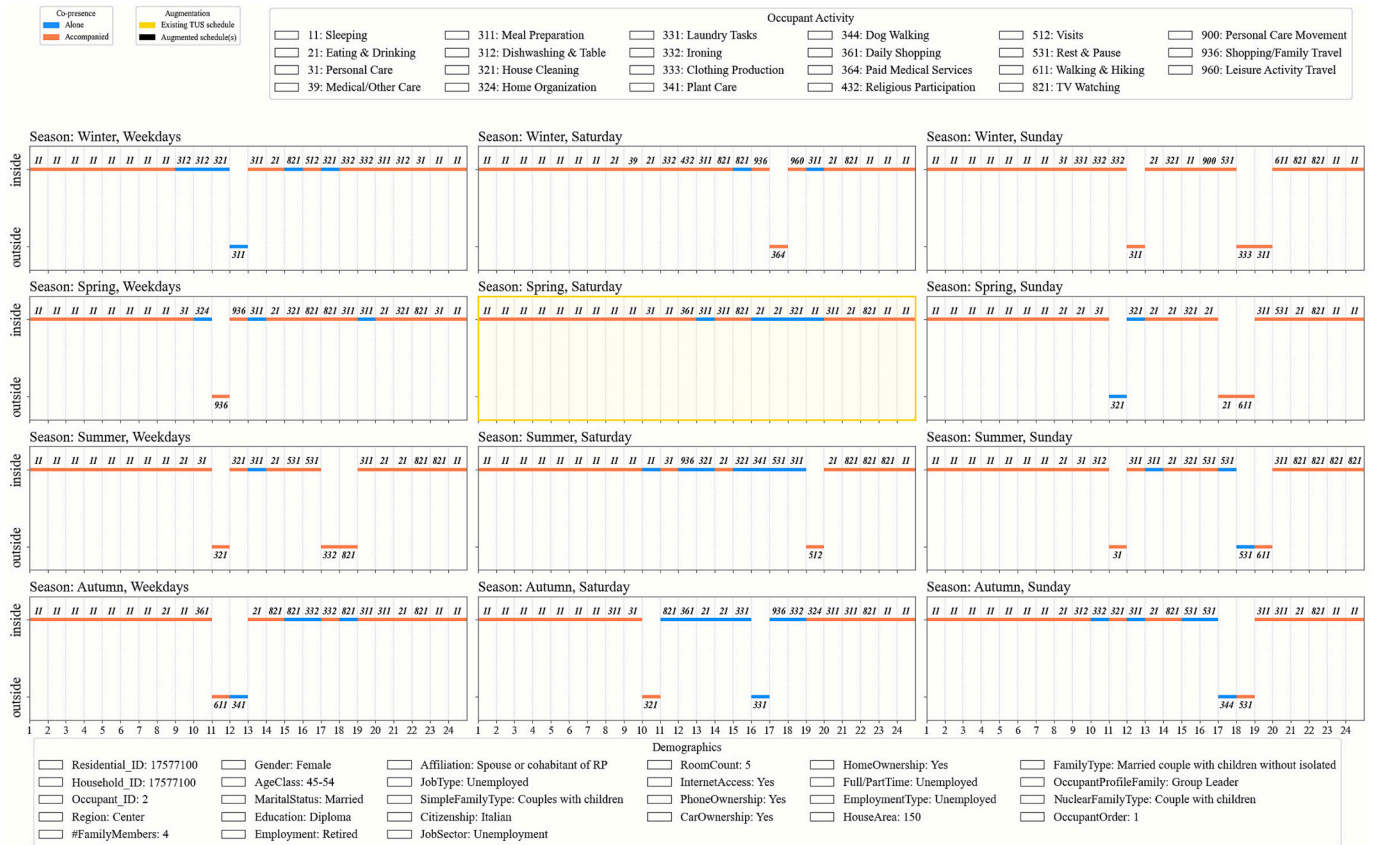


Fig. 18. Mother-Married; 4-person Household, Augmented Output Variables.

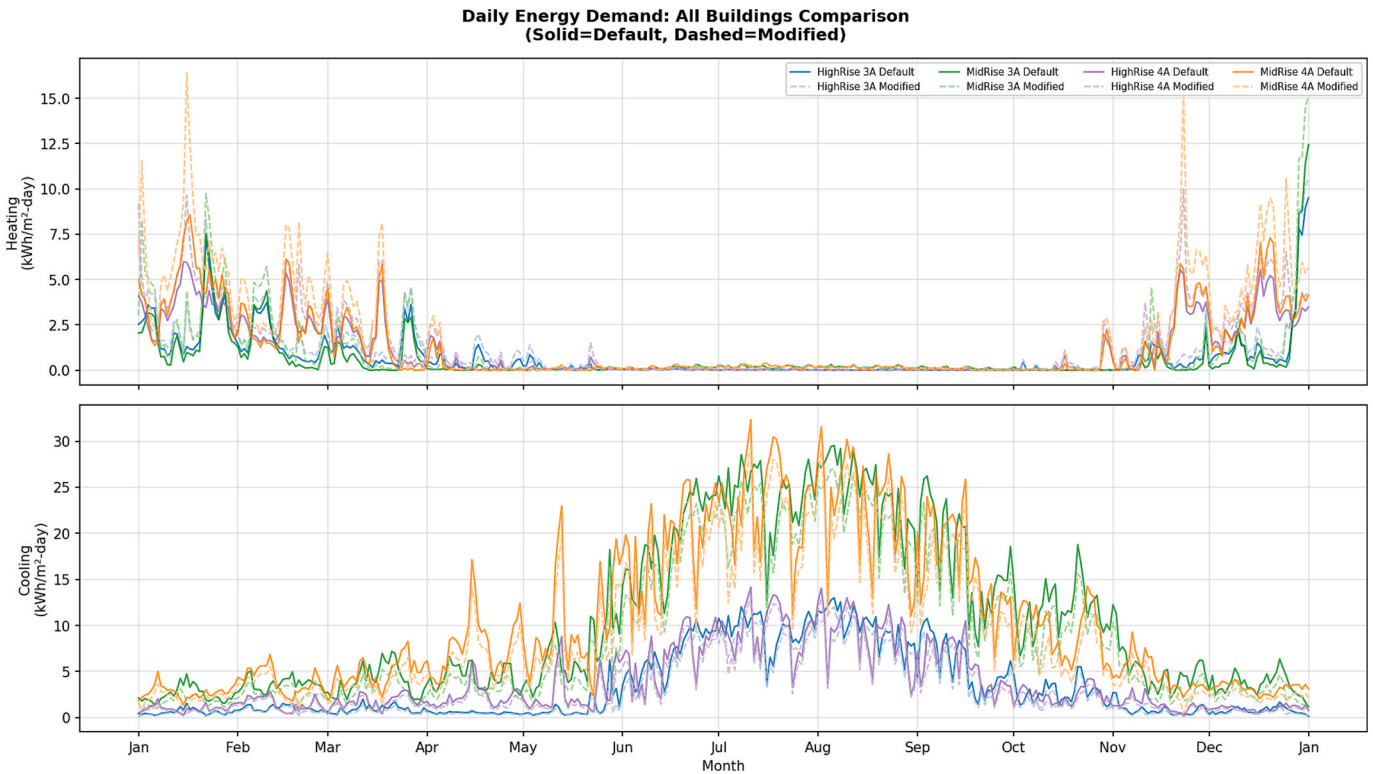


Fig. 19. Annual energy demand comparison between modified and default setups, daily resolution.

thermodynamic interactions missed by default schedule-based approaches.

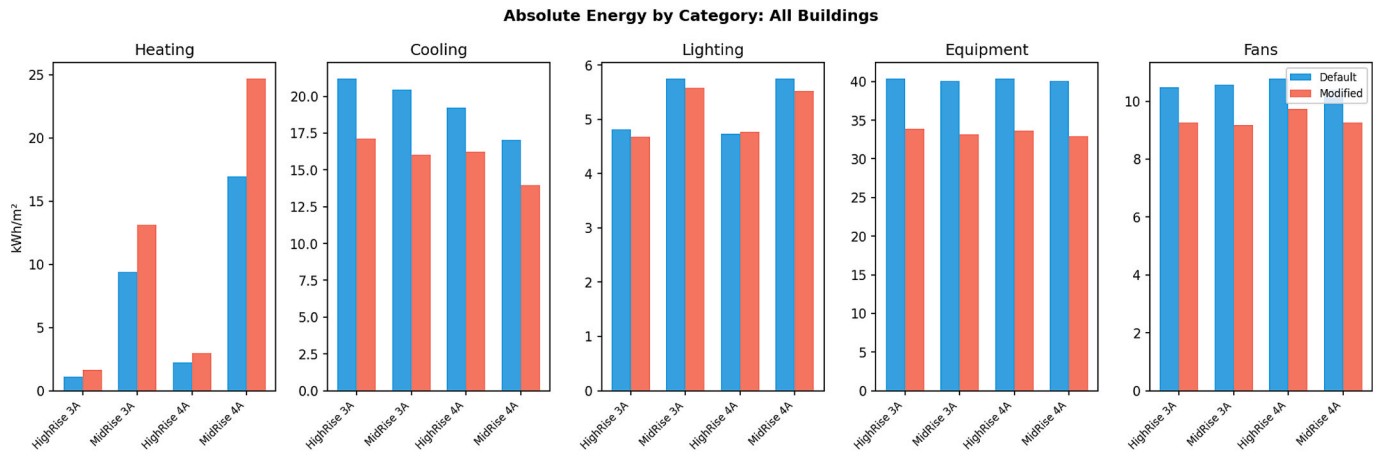


Fig. 20. Annual energy demand comparison between real vs default vs modified setups.

5. Discussion

This study shows that combining CENSUS with occupancy daily sets extends the model's applicability across regions. However, data reliability and transferability remain concerns. Integrating CENSUS with TUS is complicated by mismatched identifiers, and our privacy-preserving fusion remains an ad hoc solution. Several demographic attributes present in the TUS-Individual but absent from the CENSUS (e.g., Family Typology) had to be inferred using a multi-output neural network. This dependency means that datasets must share similar columns and categories to be integrated effectively. In summary, these discrepancies and missing attributes stem from the fact that institutional datasets were not originally designed for occupancy modeling. So, unless future surveys adopt compatible formats, researchers will continue to face extensive preprocessing challenges despite tailored protocols.

High data quality is fundamental for reliable occupancy modeling because noisy or contradictory records can propagate errors throughout the entire modeling pipeline. Large-scale datasets, such as those for Italy, also contain inconsistencies that require systematic correction, for example mismatches between household size and occupant counts. The original TUS illustrate this challenge clearly, as contradictions in "Co-Presence" responses undermined reliability. However, we created consistent daily sets for occupancy modeling with FE methods. This improved usability but reduced behavioral detail.

Experiments with different DL configurations revealed that categorical embeddings are decisive for performance. Without them, both LSTM and Transformer models failed to capture the joint dynamics of temporal and categorical attributes. Traditional encodings can represent dataset, but embeddings expand the model's capacity by mapping demographics into continuous spaces where temporal and categorical interactions are learned more effectively. A key novelty of this work is precisely this integration of non-temporal demographic attributes with temporal behavioral patterns, which goes beyond previous studies that relied mainly on time-use surveys. This integration provides demographic context for activity sequences and significantly expands both the scale and the geographic distribution of the dataset. For this reason, embeddings are particularly valuable in occupancy modeling, though they also require large, balanced datasets to avoid biased representation.

Regarding ethical implications, while the current data fusion is an ad-hoc technical solution, it offers a significant advantage over sensor-based methods. By synthesizing anonymous population statistics (CENSUS) with disconnected time-use diaries (TUS), the framework generates realistic patterns without processing identifiable personal data, thereby mitigating re-identification risks inherent in tracking specific households.

A clear pathway exists for integrating the high-fidelity schedules from CENTUS into existing UBEW workflows. The framework's time-

series array outputs can be directly translated into UBEW-ready formats; for instance, the raw time-series can be formatted as EnergyPlus "Schedule:File" objects, a standard input for hourly schedules. Furthermore, the CENTUS dataset facilitates the generation of all key occupancy components required by UBEW tools. The 'household size' attribute, for example, directly informs occupant density (people/m²). Similarly, the classified hourly activities yield detailed occupant activity schedules (e.g., metabolic rates, equipment use), which can then be aggregated to define occupant activity density (W/m²). Lastly, individual hourly presence columns can be combined to create robust, aggregated household presence schedules.

Finally, three key limitations remain. First, the intentional generalization threshold (95% accuracy) improves model robustness but makes the classification of extremely rare activities more sensitive, leading to lower performance in these categories. Second, regarding generalizability, the trained model itself is specific to the Italian context, as occupancy behaviors are intrinsically driven by local cultural and climatic factors. Therefore, transferability lies in the methodology rather than the model weights. Applying this framework to other HETUS countries requires rigorous semantic harmonization of activity codes, highlighting a critical need for the HETUS framework to become more adaptable for diverse modeling purposes. Third, daily set diversity is ultimately bounded by the source dataset parameters. Future work will address these by creating a unified multi-national HETUS dataset and implementing Transformer-based retraining, leveraging parallel processing for large-scale applications. This expanded dataset will support full annual weekday daily set generation with enhanced temporal detail. The improved hourly outputs will integrate seamlessly with BEM/UBEW for better calibration and analysis across building scales, reducing performance gaps in BEM/UBEW.

6. Conclusion

This research presents a novel, data-driven framework for occupancy modeling in residential buildings, aimed at capturing household behaviors with greater detail and scalability than traditional approaches. By integrating non-temporal (i.e., household demographics and building attributes) and temporal attributes into a unified pipeline, the study advances the development of realistic occupancy that reflect the diversity and complexity of daily life.

We accomplished the task through the implementation of a comprehensive four-stage pipeline (Fig. 5). Leveraging ethically sourced TUS and CENSUS, **our privacy-preserving DL approach transforms population statistics into detailed temporal occupancy attributes, covering entire geographical regions.** Using DL architectures, our model consistently outperforms fixed schedule-based and stochastic methods. Crucially, the framework is transferable to any country within

the HETUS network, demonstrating a viable and scalable solution for incorporating realistic occupant dynamics into city-scale energy analyses.

The research presents four primary contributions: (i) the comprehensive integration of non-temporal attributes into temporal attribute classification through DL methodologies; (ii), the exclusive reliance on TUS, treating occupancy as static attributes and daily diaries as sequential attributes to classify/augment all occupancy components without requiring [supplementary dataset](#) sources; (iii), the inherent global applicability across all nations participating HETUS framework; (iv), occupancy modeling with various sampling options, allowing it to be applied from individual residential units to entire neighborhood.

The comparative analysis of DL architectures yielded several important insights regarding model performance characteristics. Variations in hyperparameter configurations between the two architectures influenced their respective learning trajectories. Transformer, configured with reduced learning rates and enlarged internal processing units, demonstrated oscillatory accuracy patterns resulting from frequent learning rate adjustments. In contrast, LSTM, operating with elevated learning rates and moderately sized units, maintained consistent performance improvements throughout training. Both demonstrated strong performance capabilities for frequently occurring activities, though the Transformer model exhibited marginally superior performance in classifying sparse data patterns, particularly for rare activity categories.

The proposed different occupancy augmentation approaches as single mode method selects classifications based on highest probability values, while the neighborhood-scale alternatives employ probabilistic sampling techniques (i.e. probabilistic, temperature-controlled) to introduce behavioral variability. Neighborhood approaches are valuable for concurrent simulation of multiple households, ensuring that occupants sharing similar demographic characteristics exhibit varied temporal patterns. This approach significantly enhances the realism by implementing the variation into the occupancy daily sets, as outlined in 3.3.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used *Claude.ai* to check grammar, spelling. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

Orcun Koral Iseri: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Ipek Gursel Dino:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Conceptualization. **Sinan Kalkan:** Writing – review & editing, Validation, Supervision, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Time-Use Survey, released under the Creative Commons Attribution license.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enbuild.2026.117155>.

Data availability

Data will be made available on request.

References

- [1] T. Hong, Y. Chen, X. Luo, N. Luo, S.H. Lee, Ten questions on urban building energy modeling, *Build. Environ.* 168 (2020) 106508, <https://doi.org/10.1016/j.buildenv.2019.106508>.
- [2] C.F. Reinhart, C. Cerezo Davila, Urban building energy modeling - A review of a nascent field, *Build. Environ.* 97 (2016) 196–202, <https://doi.org/10.1016/j.buildenv.2015.12.001>.
- [3] U. Ali, M.H. Shamsi, C. Hoare, E. Mangina, J. O'Donnell, Review of urban building energy modeling (UBEM) approaches, methods and tools using qualitative and quantitative analysis, *Energy Build.* 246 (2021) 111073, <https://doi.org/10.1016/j.enbuild.2021.111073>.
- [4] H. Yoshino, T. Hong, N. Nord, X. Zhang, IEA EBC annex 53: Total energy use in buildings—Analysis and evaluation methods, *Energy Build.* 152 (2017) 124–136, <https://doi.org/10.1016/j.enbuild.2017.07.038>.
- [5] J. Yang, M. Santamouris, S.E. Lee, Review of occupancy sensing systems and occupancy modeling methodologies for the application in institutional buildings, *Energy Build.* (2016), <https://doi.org/10.1016/j.enbuild.2015.12.019>.
- [6] O. Ahmed, N. Sezer, M. Ouf, L. (Leon) Wang, I.G. Hassan, State-of-the-art review of occupant behavior modeling and implementation in building performance simulation, *Renewable and Sustainable Energy Reviews* 185 (2023) 113558, <https://doi.org/10.1016/j.rser.2023.113558>.
- [7] S. Zhang, Q. Guo, R. Smyth, Y. Yao, Extreme temperatures and residential electricity consumption: Evidence from Chinese households, *Energy Econ.* 107 (2022) 105890, <https://doi.org/10.1016/j.eneco.2022.105890>.
- [8] T. Hong, D. Yan, S. D'Oca, C. fei Chen, Ten questions concerning occupant behavior in buildings: The big picture, *Build. Environ.* 114 (2017) 518–530, <https://doi.org/10.1016/j.buildenv.2016.12.006>.
- [9] J. Torriti, Understanding the timing of energy demand through time use data: Time of the day dependence of social practices, *Energy Res. Soc. Sci.* 25 (2017), <https://doi.org/10.1016/j.erss.2016.12.004>.
- [10] N. van Bommel, J.I. Höffken, The urgency of climate action and the aim for justice in energy transitions – dynamics and complexity, *Environ. Innov. Soc. Transit.* 48 (2023) 100763, <https://doi.org/10.1016/j.eist.2023.100763>.
- [11] R. Melfi, B. Rosenblum, B. Nordman, K. Christensen, Measuring building occupancy using existing network infrastructure, in: *International Green Computing Conference and Workshops, IGCC 2011 (2011) 2011*, <https://doi.org/10.1109/IGCC.2011.6008560>.
- [12] T. Labeddan, W. Zeiler, G. Boxem, Y. Zhao, Occupancy measurement in commercial office buildings for demand-driven control applications - A survey and detection system evaluation, *Energy Build.* (2015), <https://doi.org/10.1016/j.enbuild.2015.02.028>.
- [13] D. Yan, W. O'Brien, T. Hong, X. Feng, H. Burak Gunay, F. Tahmasebi, A. Mahdavi, Occupant behavior modeling for building performance simulation: Current state and future challenges, *Energy Build.* 107 (2015) 264–278, <https://doi.org/10.1016/j.enbuild.2015.08.032>.
- [14] C. Carpino, D. Mora, N. Arcuri, M. De Simone, Behavioral variables and occupancy patterns in the design and modeling of Nearly Zero Energy Buildings, in: *Build. Simul., Springer*, 2017: pp. 875–888.
- [15] H.P. Tuniki, A. Jurelionis, P. Fokaides, A review on the approaches in analysing energy-related occupant behaviour research, *Journal of Building Engineering* 40 (2021) 102630, <https://doi.org/10.1016/j.jobbe.2021.102630>.
- [16] H.B. Gunay, W. O'Brien, I. Beausoleil-Morrison, A critical review of observation studies, modeling, and simulation of adaptive occupant behaviors in offices, *Build. Environ.* 70 (2013) 31–47, <https://doi.org/10.1016/j.buildenv.2013.07.020>.
- [17] S. Chen, G. Zhang, X. Xia, Y. Chen, S. Setunge, L. Shi, The impacts of occupant behavior on building energy consumption: A review, *Sustainable Energy Technologies and Assessments* 45 (2021) 101212, <https://doi.org/10.1016/j.seta.2021.101212>.
- [18] V. Fabi, R.K. Andersen, S. Corngati, Verification of stochastic behavioural models of occupants' interactions with windows in residential buildings, *Build. Environ.* 94 (2015), <https://doi.org/10.1016/j.buildenv.2015.08.016>.
- [19] M. Mosteiro-Romero, I. Hischer, J.A. Fonseca, A. Schlueter, A novel population-based occupancy modeling approach for district-scale simulations compared to standard-based methods, *Build. Environ.* 181 (2020) 107084, <https://doi.org/10.1016/j.buildenv.2020.107084>.
- [20] D. Yan, T. Hong, B. Dong, A. Mahdavi, S. D'Oca, I. Gaetani, X. Feng, IEA EBC Annex 66: Definition and simulation of occupant behavior in buildings, *Energy Build.* 156 (2017) 258–270, <https://doi.org/10.1016/j.enbuild.2017.09.084>.

- [21] R. Baetens, D. Saelens, Modelling uncertainty in district energy simulations by stochastic residential occupant behaviour, *J. Build. Perform. Simul.* (2016), <https://doi.org/10.1080/19401493.2015.1070203>.
- [22] M. Picco, R. Lollini, M. Marengo, Towards energy performance evaluation in early stage building design: A simplification methodology for commercial building models, *Energy Build.* 76 (2014) 497–505.
- [23] I. Bosu, H. Mahmoud, H. Hassan, Energy audit, techno-economic, and environmental assessment of integrating solar technologies for energy management in a university residential building: A case study, *Appl. Energy* 341 (2023) 121141, <https://doi.org/10.1016/j.apenergy.2023.121141>.
- [24] X. Zhou, Y. Lu, S. Hu, Z. Yang, D. Yan, New perspectives on temporal changes in occupancy characteristics of residential buildings, *Journal of Building Engineering* 64 (2023) 105590, <https://doi.org/10.1016/j.job.2022.105590>.
- [25] A. Heydarian, C. McIlvennie, L. Arpan, What drives our behaviors in buildings? A review on occupant interactions with building systems from the lens of behavioral theories, *Build. Environ.* 179 (2020).
- [26] J. Langevin, J. Wen, P.L. Gurian, Simulating the human-building interaction: Development and validation of an agent-based model of office occupant behaviors, *Build. Environ.* 88 (2015), <https://doi.org/10.1016/j.buildenv.2014.11.037>.
- [27] K. Akkaya, I. Guvenc, R. Aygun, N. Pala, A. Kadri, IoT-based Occupancy Monitoring Techniques for Energy-Efficient Smart, Buildings (2015) 58–63.
- [28] S. Petersen, T.H. Pedersen, K.U. Nielsen, M.D. Knudsen, Establishing an image-based ground truth for validation of sensor data-based room occupancy detection, *Energy Build.* 130 (2016), <https://doi.org/10.1016/j.enbuild.2016.09.009>.
- [29] J.Y. Park, T. Dougherty, Z.N. Intelligent, A bluetooth based occupancy detection for buildings, in: *ASHRAE and IBPSA-USA Building Simulation Conference* (2018).
- [30] J.R. Padilla-López, A.A. Chaaraoui, F. Flórez-Revuelta, Visual privacy protection methods: A survey, *Expert Syst. Appl.* 42 (2015) 4177–4195, <https://doi.org/10.1016/j.eswa.2015.01.041>.
- [31] S. Dabirian, K. Panchabikesan, U. Eicker, Occupant-centric Modeling for Urban Building Energy Simulation: Approaches, Inputs, and Data Sources - A Review, *Energy Build.* 257 (2021) 111809, <https://doi.org/10.1016/j.enbuild.2021.111809>.
- [32] G. Happle, J.A. Fonseca, A. Schlueter, A review on occupant behavior in urban building energy models, *Energy Build.* 174 (2018), <https://doi.org/10.1016/j.enbuild.2018.06.030>.
- [33] C. Duarte, R. Budwig, K. Van Den Wymelenberg, Energy and demand implication of using recommended practice occupancy diversity factors compared to real occupancy data in whole building energy simulation, *J. Build. Perform. Simul.* 8 (2015), <https://doi.org/10.1080/19401493.2014.966275>.
- [34] V. Grimm, U. Berger, F. Bastiansen, S. Eliassen, V. Ginot, J. Giske, J. Goss-Custard, T. Grand, S.K. Heinz, G. Huse, A. Huth, J.U. Jepsen, C. Jørgensen, W.M. Mooij, B. Müller, G. Pe'er, C. Piou, S.F. Railsback, A.M. Robbins, M.M. Robbins, E. Rossmanith, N. Rüger, E. Strand, S. Souissi, R.A. Stillman, R. Vabø, U. Visser, D.L. DeAngelis, A standard protocol for describing individual-based and agent-based models, *Ecol. Modell.* 198 (2006), <https://doi.org/10.1016/j.ecolmodel.2006.04.023>.
- [35] U. Wilke, F. Haldi, J.L. Scartezzini, D. Robinson, A bottom-up stochastic model to predict building occupants' time-dependent activities, *Build. Environ.* 60 (2013), <https://doi.org/10.1016/j.buildenv.2012.10.021>.
- [36] E. Azar, C.C. Menassa, Agent-Based Modeling of Occupants and Their Impact on Energy Use in Commercial, Buildings (2012), [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000158](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000158).
- [37] V.L. Erickson, M.Á. Carreira-Perpiñán, A.E. Cerpa, Observe., Occupancy-based system for efficient reduction of HVAC energy, in: *Proceedings of the 10th ACM/IEEE International Conference on Information Processing in Sensor Networks*, 2011.
- [38] G. Buttitta, W.J.N. Turner, O. Neu, D.P. Finn, Development of occupancy-integrated archetypes: Use of data mining clustering techniques to embed occupant behaviour profiles in archetypes, *Energy Build.* 198 (2019) 84–99, <https://doi.org/10.1016/j.enbuild.2019.05.056>.
- [39] S. D'Oca, T. Hong, Occupancy schedules learning process through a data mining framework, *Energy Build.* 88 (2015), <https://doi.org/10.1016/j.enbuild.2014.11.065>.
- [40] J. Chen, R. Adhikari, E. Wilson, J. Robertson, A. Fontanini, B. Polly, O. Olawale, Stochastic simulation of residential building occupant-driven energy use in a bottom-up model of the U.S. housing stock, (2021).
- [41] W. O'Brien, A. Wagner, M. Schweiker, A. Mahdavi, J. Day, M.B. Kjærgaard, S. Carlucci, B. Dong, F. Tahmasebi, D. Yan, T. Hong, H.B. Gunay, Z. Nagy, C. Miller, C. Berger, Introducing IEA EBC annex 79: Key challenges and opportunities in the field of occupant-centric building design and operation, *Build. Environ.* 178 (2020) 106738, <https://doi.org/10.1016/j.buildenv.2020.106738>.
- [42] Y. Chen, T. Hong, X. Luo, An agent-based stochastic Occupancy Simulator, *Build. Simul.* 11 (2018), <https://doi.org/10.1007/s12273-017-0379-7>.
- [43] C. Liao, Y. Lin, P. Barooah, Agent-based and graphical modelling of building occupancy, *J. Build. Perform. Simul.* 5 (2012) 5–25, <https://doi.org/10.1080/19401493.2010.531143>.
- [44] C.J. Andrews, D. Yi, U. Krogmann, J.A. Senick, R.E. Wener, Designing buildings for real occupants: An agent-based approach, *IEEE Transactions on Systems, Man, and Cybernetics Part A: Systems and Humans* 41 (2011), <https://doi.org/10.1109/TSMCA.2011.2116116>.
- [45] M. Migliori, H. Najafi, A. Fabregas, T. Nguyen, Neural Network-Based Building Energy Models for Adapting to Post-Occupancy Conditions: A Case Study for Florida, *Journal of Engineering for Sustainable Buildings and Cities* 3 (2022), <https://doi.org/10.1115/1.4056393>.
- [46] M. Wang, X. Wang, G. Zhang, C. Li, Occupancy detection based on Spiking Neural Networks for green building automation systems, in: *Proceedings of the World Congress on Intelligent Control and Automation (WCICA)*, 2015, <https://doi.org/10.1109/WCICA.2014.7053149>.
- [47] C. Zhuang, R. Choudhary, A. Mavrogianni, Probabilistic occupancy forecasting for risk-aware optimal ventilation through autoencoder Bayesian deep neural networks, *Build. Environ.* 219 (2022), <https://doi.org/10.1016/j.buildenv.2022.109207>.
- [48] B. Yang, F. Haghighat, B.C.M. Fung, K. Panchabikesan, Season-Based Occupancy Prediction in Residential Buildings Using Machine Learning Models, *E-Prime - Advances in Electrical Engineering, Electronics and Energy* 1 (2021), <https://doi.org/10.1016/j.eprime.2021.100003>.
- [49] P. Anand, C. Deb, K. Yan, J. Yang, D. Cheong, C. Sekhar, Occupancy-based energy consumption modelling using machine learning algorithms for institutional buildings, *Energy Build.* 252 (2021), <https://doi.org/10.1016/j.enbuild.2021.111478>.
- [50] B. Huchuk, S. Sanner, W. O'Brien, Comparison of machine learning models for occupancy prediction in residential buildings using connected thermostat data, *Build. Environ.* 160 (2019), <https://doi.org/10.1016/j.buildenv.2019.106177>.
- [51] S. Nejadshamsi, U. Eicker, C. Wang, J. Bentahar, Data sources and approaches for building occupancy profiles at the urban scale – A review, *Build. Environ.* 238 (2023) 110375, <https://doi.org/10.1016/j.buildenv.2023.110375>.
- [52] M. Osman, M. Ouf, A comprehensive review of time use surveys in modelling occupant presence and behavior: Data, methods, and applications, *Build. Environ.* 196 (2021) 107785, <https://doi.org/10.1016/j.buildenv.2021.107785>.
- [53] B.J. Johnson, M.R. Starke, O.A. Abdelaziz, R.K. Jackson, L.M. Tolbert, A method for modeling household occupant behavior to simulate residential energy consumption, in: *ISGT 2014* (2014) 1–5, <https://doi.org/10.1109/ISGT.2014.6816483>.
- [54] A. Sekar, E. Williams, R. Chen, Changes in Time Use and Their Effect on Energy Consumption in the United States, *Joule* 2 (2018) 521–536, <https://doi.org/10.1016/j.joule.2018.01.003>.
- [55] . Anderson, J. Torriti, Explaining shifts in UK electricity demand using time use data from 1974 to 2014, *Energy Policy* 123 (2018) 544–557, <https://doi.org/10.1016/j.enpol.2018.09.025>.
- [56] S. Vosoughkhosravi, A. Jafari, Y. Zhu, Application of American time use survey (ATUS) in modelling energy-related occupant-building interactions: A comprehensive review, *Energy Build.* 294 (2023) 113245, <https://doi.org/10.1016/j.enbuild.2023.113245>.
- [57] ISTAT, 15th Population and housing census 2011, Rome, Italy, 2013.
- [58] Istat, Multiscopo ISTAT – Time Use, (2013).